Calculus

Last modified: 2026-10-13 18:25:48 (PDT)

Derivatives

Limits and derivatives

Definition 0.1 (Limit of a function at a point)

Definition 0.1 (Limit of a function at a point)**.** Let f be a function defined at every point of an open interval around c, except possibly at c itself, and let L be a real number. The **limit** of $f(x)$ as x approaches c is L, written $\lim_{x \to c} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{for every } x \text{ with } 0 < |x - c| < \delta.$$

Here $|\cdot|$ is the absolute value. When such a real number L exists, the limit **exists**; otherwise, the limit does not exist. The value $f(c)$, if it is defined, plays no role.

Example 0.1 (The limit of $3x + 1$ at 2)

Example 0.1 (The limit of $3x + 1$ at 2)**.** $\lim_{x \to 2}(3x + 1) = 7$. Given $\varepsilon > 0$, take $\delta = \varepsilon/3$. For every x with $0 < |x - 2| < \delta$, using the distributive law in the second line,

$$\begin{aligned} |(3x + 1) - 7| &= |3x - 6| && \text{(subtract)} \\ &= |3(x - 2)| && \text{(distributive law)} \\ &= 3|x - 2| && (|3y| = 3|y|, \text{ since } 3 > 0) \\ &< 3 \cdot \frac{\varepsilon}{3} && (|x - 2| < \delta = \varepsilon/3) \\ &= \varepsilon && \text{(multiply)} \end{aligned}$$

For example, with $\varepsilon = 0.3$ and $\delta = 0.1$, the point $x = 2.05$ satisfies $0 < |2.05 - 2| < 0.1$, and

$$\begin{aligned} |(3 \cdot 2.05 + 1) - 7| &= |7.15 - 7| \\ &= 0.15 \\ &< 0.3. \end{aligned}$$

Definition 0.2 (One-sided limits)

Definition 0.2 (One-sided limits)**.** Let f be a function and L a real number.

- The **right-hand limit** of f at c is L, written $\lim_{x \to c^+} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for every x with $c < x < c + \delta$.
- The **left-hand limit** of f at c is L, written $\lim_{x \to c^-} f(x) = L$, if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for every x with $c - \delta < x < c$.

These are the **one-sided limits** of f at c. The limit $\lim_{x \to c} f(x)$ (Definition 0.1) exists exactly when both one-sided limits exist and are equal, and then all three are equal.

Example 0.2 (One-sided limits of a step function)

Example 0.2 (One-sided limits of a step function)**.** Let $H(x) = 1$ for $x \geq 0$ and $H(x) = 0$ for $x < 0$.

- $\lim_{x \to 0^+} H(x) = 1$, because $H(x) = 1$ for every $x > 0$.
- $\lim_{x \to 0^-} H(x) = 0$, because $H(x) = 0$ for every $x < 0$.

The one-sided limits differ, since $1 \neq 0$, so $\lim_{x \to 0} H(x)$ does not exist.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist introduces limits and derivatives:

- Calculating Limits
- Exercises on Limits
- Intro to Differential Calculus
- How Derivatives Arise from Limits
- Derivative Notation

Definition 0.3 (Difference quotient)

Definition 0.3 (Difference quotient)**.** Let f be a real-valued function of one variable, defined at c and at $c + h$, with $h \neq 0$. The **difference quotient** of f at c with increment

h is

$$\frac{f(c+h)-f(c)}{h},$$

the slope of the line through the points $(c, f(c))$ and $(c+h, f(c+h))$.

Example 0.3 (The difference quotient of x^2 at 1)

Example 0.3 (The difference quotient of x^2 at 1)**.** For $f(x) = x^2$, $c = 1$, and $h \neq 0$,

$$\begin{aligned}
\frac{f(1+h)-f(1)}{h} &= \frac{(1+h)^2-1}{h} && \text{(substitute into } f\text{)} \\
&= \frac{1+2h+h^2-1}{h} && \text{(expand } (1+h)^2\text{)} \\
&= \frac{2h+h^2}{h} && (1-1=0) \\
&= 2+h, && \text{(divide by } h \neq 0\text{)}
\end{aligned}$$

which tends to 2 as $h \to 0$ (Definition 0.1). With $h = 0.1$ the difference quotient is $2 + 0.1 = 2.1$: the line through $(1, 1)$ and $(1.1, 1.21)$ has slope

$$\begin{aligned}
\tfrac{1.21-1}{0.1} &= \tfrac{0.21}{0.1} \\
&= 2.1.
\end{aligned}$$

Definition 0.4 (Differentiable function)

Definition 0.4 (Differentiable function)**.** A function f is **differentiable at** $x = c$ if the limit (Definition 0.1) of the difference quotient (Definition 0.3) as $h \to 0$,

$$f'(c) \stackrel{\text{def}}{=} \lim_{h \to 0} \frac{f(c+h)-f(c)}{h},$$

exists and is finite. The number $f'(c)$ is the derivative of f at c (Definition 0.5). (Larson and Edwards 2018, sec. 2.1, p. 100)

Example 0.4 (A differentiable function, and a limit that is not finite)

Example 0.4 (A differentiable function, and a limit that is not finite)**.**

- $f(x) = x^2$ is differentiable at $c = 3$:

$$\begin{aligned}
\frac{f(3+h)-f(3)}{h} &= \frac{(3+h)^2-3^2}{h} && \text{(substitute } f(x)=x^2\text{)} \\
&= \frac{9+6h+h^2-9}{h} && \text{(expand } (3+h)^2\text{)} \\
&= \frac{6h+h^2}{h} && \text{(cancel } 9-9\text{)} \\
&= 6+h && \text{(divide by } h \neq 0\text{)}
\end{aligned}$$

which tends to the finite limit $f'(3) = 6$ as $h \to 0$.

- $g(x) = \sqrt[3]{x}$ is not differentiable at $c = 0$: the difference quotient

$$\begin{aligned}
\frac{g(h)-g(0)}{h} &= \frac{\sqrt[3]{h}}{h} \\
&= \frac{1}{(\sqrt[3]{h})^2}
\end{aligned}$$

grows without bound as $h \to 0$, so the limit is not finite.

Definition 0.5 (Derivative)

Definition 0.5 (Derivative)**.** If f is differentiable at c (Definition 0.4), the number $f'(c)$ is the **derivative** of f at c. The **derivative** of f is the function f' whose value at each x where f is differentiable is $f'(x)$. Writing the derivative with a prime, as f' or $f'(x)$, is **Lagrange's notation**. The derivative of f is also written $\frac{df}{dx}$ or $\frac{d}{dx}f(x)$, which is **Leibniz notation**.

Example 0.5 (The derivative of x^2)

Example 0.5 (The derivative of x^2)**.** For $f(x) = x^2$, the computation in Example 0.4, with 3 replaced by any real number c, gives $\frac{f(c+h)-f(c)}{h} = 2c + h$, which tends to $2c$ as $h \to 0$. So the derivative of f is $f'(x) = 2x$. For example, $f'(3) = 6$ and $f'(-1) = -2$.

Definition 0.6 (One-sided derivatives)

Definition 0.6 (One-sided derivatives)**.** Let f be a function defined at c.

- The **right-hand derivative** of f at c is the right-hand limit (Definition 0.2) $\lim_{h\to 0^+} \frac{f(c+h)-f(c)}{h}$ of the difference quotient (Definition 0.3), when that limit exists and is finite.
- The **left-hand derivative** of f at c is the left-hand limit $\lim_{h\to 0^-} \frac{f(c+h)-f(c)}{h}$, when that limit exists and is finite.

These are the **one-sided derivatives** of f at c.

Example 0.6 (One-sided derivatives of $|x|$ at 0)

Example 0.6 (One-sided derivatives of $|x|$ at 0)**.** Let $f(x) = |x|$, the absolute value, and $c = 0$.

- For $h > 0$, $|h| = h$, so

$$\begin{aligned}\frac{f(0+h) - f(0)}{h} &= \frac{h - 0}{h} \\ &= 1.\end{aligned}$$

  The right-hand derivative of f at 0 is 1.

- For $h < 0$, $|h| = -h$, so

$$\begin{aligned}\frac{f(0+h) - f(0)}{h} &= \frac{-h - 0}{h} \\ &= -1.\end{aligned}$$

  The left-hand derivative of f at 0 is -1.

For example, $h = 0.5$ gives $\frac{0.5}{0.5} = 1$, and $h = -0.5$ gives $\frac{0.5}{-0.5} = -1$. The one-sided derivatives differ, since $1 \neq -1$.

Definition 0.7 (Second derivative)

Definition 0.7 (Second derivative)**.** Let f be a real-valued function of one variable whose derivative f' (Definition 0.5) exists at every point of an open interval containing c. If f' is differentiable (Definition 0.4) at c, its derivative there, written $f''(c)$ or $\frac{d^2 f}{dx^2}$, is the **second derivative** of f at c.

Example 0.7 (The second derivative of x^3)

Example 0.7 (The second derivative of x^3)**.** For $f(x) = x^3$, the power rule (Theorem 0.3) gives $f'(x) = 3x^2$ at every x, and differentiating again, with the constant multiple rule (Theorem 0.2), gives

$$\begin{aligned}f''(x) &= 3 \cdot 2x \\ &= 6x.\end{aligned}$$

So $f''(2) = 12$, $f''(0) = 0$, and $f''(-1) = -6$.

Derivative rules

Theorem 0.1 (Constant rule)

Theorem 0.1 (Constant rule)**.** *If c is constant with respect to x, then*

$$\frac{\partial}{\partial x}c = 0$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Derivative of a Constant

Theorem 0.2 (Constant multiple rule)

Theorem 0.2 (Constant multiple rule)**.** *If a is constant with respect to x and y is a differentiable function of x, then:*

$$\frac{\partial}{\partial x}ay = a\frac{\partial y}{\partial x}$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Constant Multiple Rule for Derivatives

Theorem 0.3 (Power rule)

Theorem 0.3 (Power rule)**.** *For every real number q and every $x > 0$, the derivative of the power x^q is:*

$$\frac{\partial}{\partial x}x^q = qx^{q-1}$$

When q is a positive integer, the same formula holds for every real x.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on this rule, on the sum rule (the derivative of a sum is the sum of the derivatives), and on exercises

that combine the rules so far:

- The Power Rule for Derivatives
- The Sum Rule for Derivatives
- Exercises on Derivative Rules

Exercise 0.1 (Derivative of polynomial and power functions)

Exercise 0.1 (Derivative of polynomial and power functions)**.** Adapted from Miller (2016), Question 1.1.1.
Find the derivative of

$$f(x) = 4x^5 + 3x^2 + x^{1/3}$$

Solution 0.1

Solution 0.1. Differentiate term by term using the constant multiple rule (Theorem 0.2) and the power rule (Theorem 0.3):

$$\begin{aligned}
\frac{d}{dx}(4x^5) &= 4 \cdot (5x^4) \\
&= 20x^4 \\
\frac{d}{dx}(3x^2) &= 3 \cdot (2x) \\
&= 6x \\
\frac{d}{dx}(x^{1/3}) &= \frac{1}{3}x^{1/3-1} \\
&= \frac{1}{3}x^{-2/3}
\end{aligned}$$

Combining the three terms yields:

$$f'(x) = 20x^4 + 6x + \frac{1}{3}x^{-2/3}$$

Exercise 0.2 (Finding points with a prescribed slope)

Exercise 0.2 (Finding points with a prescribed slope)**.** Adapted from Miller (2016), Question 1.1.14.
Let $f(x) = \frac{1}{3}x^3 + x^2 - x - 1$. Find all points on the graph of f where the slope is:

1. -1
2. 2

3. 0

Solution 0.2

Solution 0.2. The slope is given by the derivative:

$$f'(x) = x^2 + 2x - 1$$

1. Slope -1: Solve

$$\begin{aligned} x^2 + 2x - 1 = -1 &\iff x^2 + 2x = 0 \\ &\iff x(x+2) = 0. \end{aligned}$$

The roots are $x = 0$ and $x = -2$. Evaluating f: $f(0) = -1$ and

$$\begin{aligned} f(-2) &= -\frac{8}{3} + 4 + 2 - 1 \\ &= \frac{7}{3}. \end{aligned}$$

The points are $(0, -1)$ and $(-2, 7/3)$.

2. Slope 2: Solve

$$\begin{aligned} x^2 + 2x - 1 = 2 &\iff x^2 + 2x - 3 = 0 \\ &\iff (x+3)(x-1) = 0. \end{aligned}$$

The roots are $x = -3$ and $x = 1$. Evaluating f:

$$\begin{aligned} f(-3) &= -9 + 9 + 3 - 1 \\ &= 2 \end{aligned}$$

and

$$\begin{aligned} f(1) &= \frac{1}{3} + 1 - 1 - 1 \\ &= -\frac{2}{3}. \end{aligned}$$

The points are $(-3, 2)$ and $(1, -2/3)$.

3. Slope 0: Solve $x^2 + 2x - 1 = 0$. By the quadratic formula:

$$\begin{aligned} x &= \frac{-2 \pm \sqrt{4 - 4(1)(-1)}}{2} \\ &= \frac{-2 \pm \sqrt{8}}{2} \\ &= -1 \pm \sqrt{2} \end{aligned}$$

Theorem 0.4 (Derivative of natural logarithm)

Theorem 0.4 (Derivative of natural logarithm)**.** *For every $x > 0$, the derivative of the natural logarithm is:*

$$\begin{aligned}\log'\{x\} &= \frac{1}{x} \\ &= x^{-1}\end{aligned}$$

Exercise 0.3 (Derivative after logarithmic simplification)

Exercise 0.3 (Derivative after logarithmic simplification)**.** Adapted from Miller (2016), Question 1.1.4.
Find the derivative of

$$f(x) = \log(4x) - \log(2x)$$

for $x > 0$.

Solution 0.3

Solution 0.3. Simplify before differentiating using the quotient property of logarithms:

$$\begin{aligned}f(x) &= \log(4x) - \log(2x) \\ &= \log\left(\frac{4x}{2x}\right) \\ &= \log 2\end{aligned}$$

Because $\log 2$ is constant with respect to x, the constant rule (Theorem 0.1) gives:

$$f'(x) = 0$$

Remark: While one could apply the chain rule separately to each term, giving

$$\begin{aligned}\frac{4}{4x} - \frac{2}{2x} &= \frac{1}{x} - \frac{1}{x} \\ &= 0,\end{aligned}$$

simplifying algebraically first is faster and prevents arithmetic errors.

Theorem 0.5 (Derivative of exponential)

Theorem 0.5 (Derivative of exponential)**.** *For every real x, the derivative of the exponential function is:*

$$\exp'\{x\} = \exp\{x\}$$

Theorem 0.6 (Derivative of sine)

Theorem 0.6 (Derivative of sine)**.** *For every real x, the derivative of the sine function is:*

$$\frac{\partial}{\partial x} \sin x = \cos x$$

Theorem 0.7 (Derivative of cosine)

Theorem 0.7 (Derivative of cosine)**.** *For every real x, the derivative of the cosine function is:*

$$\frac{\partial}{\partial x} \cos x = -\sin x$$

Theorem 0.8 (Product rule)

Theorem 0.8 (Product rule)**.** *If a and b are differentiable functions of x, then*

$$(ab)' = ab' + ba'$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Product Rule for Derivatives

Exercise 0.4 (Derivative of a product)

Exercise 0.4 (Derivative of a product)**.** Adapted from Miller (2016), Question 1.1.2. Find the derivative of

$$f(x) = (x^4 + 3x^2 + 8)\cos x$$

Solution 0.4

Solution 0.4. Apply the product rule (Theorem 0.8) with:

$$u(x) = x^4 + 3x^2 + 8, \qquad v(x) = \cos x$$

Their derivatives are:

$$u'(x) = 4x^3 + 6x, \qquad v'(x) = -\sin x$$

By the product rule:

$$\begin{aligned} f'(x) &= u'(x)v(x) + u(x)v'(x) \\ &= (4x^3 + 6x)\cos x - (x^4 + 3x^2 + 8)\sin x \end{aligned}$$

Exercise 0.5 (Second derivative of a product)

Exercise 0.5 (Second derivative of a product)**.** Adapted from Miller (2016), Question 1.1.15.
Find the second derivative of

$$f(x) = (x^4 + 3x^2 + 8)\cos x$$

Solution 0.5

Solution 0.5. From Exercise 0.4, the first derivative is:

$$f'(x) = (4x^3 + 6x)\cos x - (x^4 + 3x^2 + 8)\sin x$$

Differentiate each term via the product rule:

$$\begin{aligned} \frac{d}{dx}\left[(4x^3 + 6x)\cos x\right] &= (12x^2 + 6)\cos x - (4x^3 + 6x)\sin x \\ \frac{d}{dx}\left[-(x^4 + 3x^2 + 8)\sin x\right] &= -(4x^3 + 6x)\sin x - (x^4 + 3x^2 + 8)\cos x \end{aligned}$$

Combining and grouping like trigonometric terms:

$$\begin{aligned} f''(x) &= \left[(12x^2 + 6) - (x^4 + 3x^2 + 8)\right]\cos x - 2(4x^3 + 6x)\sin x \\ &= (-x^4 + 9x^2 - 2)\cos x - (8x^3 + 12x)\sin x \end{aligned}$$

Theorem 0.9 (Quotient rule)

Theorem 0.9 (Quotient rule)**.** *If a and b are differentiable functions of x and $b \neq 0$, then*

$$(a/b)' = a'/b - (a/b^2)b'$$

Video

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has a video on this rule:

- The Quotient Rule for Derivatives

Exercise 0.6 (Derivative of a rational function)

Exercise 0.6 (Derivative of a rational function)**.** Adapted from Miller (2016), Question 1.1.12.
Find the derivative of

$$f(x) = \frac{x^2 - 1}{x - 1}$$

for $x \neq 1$.

Solution 0.6

Solution 0.6. Factor the numerator before differentiating:

$$\begin{aligned} f(x) &= \frac{(x-1)(x+1)}{x-1} \\ &= x + 1 \qquad (x \neq 1) \end{aligned}$$

Differentiating directly gives $f'(x) = 1$.
Alternative (quotient rule): Using Theorem 0.9 with $u(x) = x^2 - 1$ and $v(x) = x - 1$:

$$\begin{aligned} f'(x) &= \frac{2x(x-1) - (x^2-1)(1)}{(x-1)^2} \\ &= \frac{x^2 - 2x + 1}{(x-1)^2} \\ &= \frac{(x-1)^2}{(x-1)^2} \\ &= 1 \end{aligned}$$

Both methods yield 1, but factoring first eliminates tedious algebraic simplification.

Theorem 0.10 (Chain rule)

Theorem 0.10 (Chain rule)**.** *If a is a differentiable function of b, and b is a differentiable function of c, then a is a differentiable function of c, and*

$$\begin{aligned}\frac{da}{dc} &= \frac{da}{db}\frac{db}{dc} \\ &= \frac{db}{dc}\frac{da}{db}\end{aligned}$$

or in Lagrange's notation (Definition 0.5), if g is differentiable at x and f is differentiable at $g(x)$, then for the composition $f \circ g$, with inner function g and outer function f:

$$(f(g(x)))' = g'(x)f'(g(x))$$

Exercise 0.7 (Derivative of the standard Gaussian kernel)

Exercise 0.7 (Derivative of the standard Gaussian kernel)**.** Adapted from Miller (2016), Question 1.1.5.
Find the derivative of

$$f(x) = \exp\left\{-\frac{x^2}{2}\right\}$$

Solution 0.7

Solution 0.7. Apply the chain rule (Theorem 0.10) with outer function $\exp\{u\}$ and inner function $u(x) = -x^2/2$. Since $\frac{d}{du}\exp\{u\} = \exp\{u\}$ (Theorem 0.5) and $u'(x) = -x$:

$$\begin{aligned}f'(x) &= u'(x)\exp\{u(x)\} \\ &= -x\exp\left\{-\frac{x^2}{2}\right\}\end{aligned}$$

Exercise 0.8 (Second derivative of the Gaussian kernel)

Exercise 0.8 (Second derivative of the Gaussian kernel)**.** Adapted from Miller (2016), Question 1.1.6.
Find the second derivative of

$$f(x) = \exp\left\{-\frac{x^2}{2}\right\}$$

Solution 0.8

Solution 0.8. From Exercise 0.7, the first derivative is $f'(x) = -x\exp\{-x^2/2\}$. Apply the product rule (Theorem 0.8) to $u(x) = -x$ and $v(x) = \exp\{-x^2/2\}$:

$$u'(x) = -1$$
$$v'(x) = -x \exp\left\{-\frac{x^2}{2}\right\}$$

Therefore:

$$\begin{aligned} f''(x) &= u'(x)v(x) + u(x)v'(x) \\ &= (-1)\exp\left\{-\frac{x^2}{2}\right\} + (-x)\left[-x\exp\left\{-\frac{x^2}{2}\right\}\right] \\ &= (x^2-1)\exp\left\{-\frac{x^2}{2}\right\} \end{aligned}$$

Exercise 0.9 (Combining product and chain rules)

Exercise 0.9 (Combining product and chain rules)**.** Adapted from Miller (2016), Question 1.1.7.
Find the derivative of

$$f(x) = \exp\{x^8\}\cos(3x^4)$$

Solution 0.9

Solution 0.9. Apply the product rule (Theorem 0.8) to $u(x) = \exp\{x^8\}$ and $v(x) = \cos(3x^4)$:

$$f'(x) = u'(x)v(x) + u(x)v'(x)$$

Compute $u'(x)$ and $v'(x)$ via the chain rule (Theorem 0.10):

$$u'(x) = 8x^7 \exp\{x^8\}$$
$$v'(x) = -12x^3 \sin(3x^4)$$

Substitute both derivatives into the product rule:

$$\begin{aligned} f'(x) &= 8x^7\exp\{x^8\}\cos(3x^4) - 12x^3\exp\{x^8\}\sin(3x^4) \\ &= 4x^3\exp\{x^8\}[2x^4\cos(3x^4) - 3\sin(3x^4)] \end{aligned}$$

Exercise 0.10 (Generalized power rule)

Exercise 0.10 (Generalized power rule)**.** Adapted from Miller (2016), Question 1.1.13.
Find the derivative of

$$\begin{aligned} f(x) &= \sqrt[3]{(5x-2)^2} \\ &= (5x-2)^{2/3} \end{aligned}$$

for $x \neq 2/5$.

Solution 0.10

Solution 0.10. Apply the generalized power rule $\frac{d}{dx}[g(x)]^r = r[g(x)]^{r-1}g'(x)$ with $g(x) = 5x-2$ and $r = 2/3$:

$$\begin{aligned} f'(x) &= \frac{2}{3}(5x-2)^{2/3-1} \cdot 5 \\ &= \frac{10}{3}(5x-2)^{-1/3} \\ &= \frac{10}{3(5x-2)^{1/3}} \end{aligned}$$

Remark: Omitting the inner derivative $g'(x) = 5$ is a frequent mistake when applying the generalized power rule.

Corollary 0.1 (Chain rule for logarithms)

Corollary 0.1 (Chain rule for logarithms)**.** *If f is differentiable at x and $f(x) > 0$, then*

$$\frac{d}{dx}\log\{f(x)\} = \frac{f'(x)}{f(x)}$$

Proof

Proof. Apply Theorem 0.10 and Theorem 0.4:

$$\begin{aligned} \frac{d}{dx}\log\{f(x)\} &= f'(x) \cdot \log'\{f(x)\} && \text{(chain rule, with } g = f \text{ and outer function } \log\text{)} \\ &= f'(x) \cdot \frac{1}{f(x)} && \text{(derivative of } \log\text{, valid because } f(x) > 0\text{)} \\ &= \frac{f'(x)}{f(x)} && \text{(multiply)} \end{aligned}$$

□

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on this rule and on exercises that combine it with the other rules:

- The Chain Rule for Derivatives
- The Power Rule on a Function Chain
- Advanced Exercises on Derivative Rules

Exercise 0.11 (Chain rule with logarithm)

Exercise 0.11 (Chain rule with logarithm)**.** Adapted from Miller (2016), Question 1.1.3. Find the derivative of

$$f(x) = \log(1 - x^2)$$

for $x \in (-1, 1)$.

Solution 0.11

Solution 0.11. Apply the chain rule for logarithms (Corollary 0.1): if $g(x) = 1 - x^2$, then $g'(x) = -2x$, and

$$\begin{aligned} f'(x) &= \frac{g'(x)}{g(x)} \\ &= -\frac{2x}{1 - x^2} \end{aligned}$$

Remark: A common pitfall in chain rule problems is evaluating the outer derivative at x rather than at the inner value $g(x)$. The denominator is $g(x) = 1 - x^2$, not x.

Exercise 0.12 (Second derivative with an added constant)

Exercise 0.12 (Second derivative with an added constant)**.** Adapted from Miller (2016), Question 1.1.9.
Find the second derivative of

$$f(x) = \log x + \sqrt{162}$$

for $x > 0$.

Solution 0.12

Solution 0.12. Because $\sqrt{162}$ is constant with respect to x, its derivative is zero (Theorem 0.1):

$$\begin{aligned} f'(x) &= \frac{1}{x} \\ &= x^{-1} \end{aligned}$$

Differentiating again via the power rule (Theorem 0.3):

$$\begin{aligned} f''(x) &= -x^{-2} \\ &= -\frac{1}{x^2} \end{aligned}$$

Tangent lines

Definition 0.8 (Tangent line)

Definition 0.8 (Tangent line)**.** Let f be differentiable at c (Definition 0.4). The **tangent line** to the graph of f at c is the graph of the affine function

$$x \mapsto f(c) + f'(c)\,(x - c),$$

the line through the point $(c, f(c))$ with slope $f'(c)$. The slope $f'(c)$ is also called the **tangent slope** of f at c.

Example 0.8 (The tangent line to x^2 at 1)

Example 0.8 (The tangent line to x^2 at 1)**.** For $f(x) = x^2$ at $c = 1$, $f(1) = 1$ and $f'(1) = 2$ (Example 0.5), so the tangent line is the graph of

$$\begin{aligned} x \mapsto 1 + 2\,(x - 1) \quad & \text{(substitute } f(1) = 1 \text{ and } f'(1) = 2\text{)} \\ = 2x - 1 \quad & \text{(distribute, and } 1 - 2 = -1\text{)} \end{aligned}$$

At $x = 1.1$ the tangent line has height $2(1.1) - 1 = 1.2$, close to the curve's height $f(1.1) = 1.21$; the gap, $1.21 - 1.2 = 0.01$, is $(1.1 - 1)^2$.

Linear approximation

For a differentiable function f and a small step ε,

$$f(w+\varepsilon) \approx f(w) + \varepsilon \frac{d}{dw} f(w) \tag{1}$$

Definition 0.9 (Linear approximation (first-order Taylor approximation))

Definition 0.9 (Linear approximation (first-order Taylor approximation))**.** For a function f that is differentiable at w, the **linear approximation** of f at w (also called the **first-order Taylor approximation**) is the function of the step ε

$$\hat{f}_w(\varepsilon) = f(w) + \varepsilon \frac{d}{dw} f(w)$$

so Equation 1 says $f(w+\varepsilon) \approx \hat{f}_w(\varepsilon)$ for small ε. Figure 1 illustrates this approximation interactively for a quadratic function.

Remark 0.1 (The "linear" approximation is affine)

Remark 0.1 (The "linear" approximation is affine). The linear approximation $\hat{f}_w$ (Definition 0.9) is an affine function of the step ε: its slope is $\frac{d}{dw} f(w)$ and its intercept is $f(w)$. It is linear in ε only when $f(w) = 0$. The name "linear approximation" uses "linear" in the looser sense of elementary algebra (remark). Boyd and Vandenberghe (2018), section 2.2, treats the first-order Taylor approximation as an affine function for functions of several variables as well.

(This plot is interactive; see the HTML or slide version of this page to view it.)

Figure 1: The linear approximation Equation 1 for $f(w) = w^2 - 4w + 7$, at any w and step ε.

Exercise 0.13 (Tangent line approximation)

Exercise 0.13 (Tangent line approximation)**.** Adapted from Miller (2016), Question 1.1.8.
Let $f(x) = 4x + \sqrt{2}\cos x$.

1. Compute $f'(x)$.
2. Find the equation of the tangent line to the curve $y = f(x)$ at $x = \pi/4$.
3. Use the tangent line to approximate $f(\pi/4 + 0.01)$, and compare that approximation with the exact value.

Solution 0.13

Solution 0.13. **1.** Differentiate term by term:

$$f'(x) = 4 - \sqrt{2}\sin x$$

2. Evaluate f and f' at $x_0 = \pi/4$:

$$\begin{aligned} f(\pi/4) &= 4\left(\frac{\pi}{4}\right) + \sqrt{2}\cos\left(\frac{\pi}{4}\right) \\ &= \pi + \sqrt{2}\left(\frac{\sqrt{2}}{2}\right) \\ &= \pi + 1 \\ f'(\pi/4) &= 4 - \sqrt{2}\sin\left(\frac{\pi}{4}\right) \\ &= 4 - \sqrt{2}\left(\frac{\sqrt{2}}{2}\right) \\ &= 4 - 1 \\ &= 3 \end{aligned}$$

Using the point-slope formula, the tangent line at $(\pi/4, \pi + 1)$ is:

$$y - (\pi + 1) = 3\left(x - \frac{\pi}{4}\right) \implies y = \pi + 1 + 3\left(x - \frac{\pi}{4}\right)$$

3. At $x = \pi/4 + 0.01$, the linear approximation (Equation 1) gives:

$$f(\pi/4 + 0.01) \approx (\pi + 1) + 3(0.01) = \pi + 1.03 \approx 4.171593$$

The exact value is:

$$f(\pi/4 + 0.01) = 4\left(\frac{\pi}{4} + 0.01\right) + \sqrt{2}\cos\left(\frac{\pi}{4} + 0.01\right) \approx 4.171543$$

The approximation error is $|4.171593 - 4.171543| \approx 0.00005$, on the order of $(0.01)^2 = 10^{-4}$.

Critical points and optimization

Definition 0.10 (Flat point (stationary point))

Definition 0.10 (Flat point (stationary point))**.** Let f be differentiable at c (Definition 0.4). If $f'(c) = 0$, then c is a **flat point** of f (also called a **stationary point**).

Example 0.9 (Flat points, and a point that is not one)

Example 0.9 (Flat points, and a point that is not one)**.**

- $f(w) = w^3$ has $f'(w) = 3w^2$, which is 0 at $w = 0$, so 0 is a flat point. Yet $f(0) = 0$ is neither the minimum nor the maximum of the values f takes on any open interval around 0: $f(w) < 0$ for $w < 0$ and $f(w) > 0$ for $w > 0$.
- $h(w) = w^2$ has $h'(1) = 2 \neq 0$, so 1 is not a flat point.

Exercise 0.14 (Find the flat point, and check the approximation)

Exercise 0.14 (Find the flat point, and check the approximation)**.** Let $f(w) = w^2 - 4w + 7$.

1. Differentiate f.
2. Find the flat point w of f, and say whether $f(w)$ is the minimum or the maximum of the values of f.
3. Evaluate the derivative at $w = 1$, use Equation 1 to predict $f(1.01)$, and compare that prediction with the exact value.

Solution 0.14

Solution 0.14. **1.** Term by term:

$$\frac{df}{dw} = 2w - 4$$

2. Set it to zero: $2w - 4 = 0$ gives $w = 2$.

$$\begin{aligned} f(2) &= 4 - 8 + 7 \\ &= 3 \end{aligned}$$

is the minimum of the values of f. Completing the square, $f(w) = (w-2)^2 + 3$, since

$$\begin{aligned} (w-2)^2 + 3 &= w^2 - 4w + 4 + 3 \\ &= w^2 - 4w + 7, \end{aligned}$$

and $(w-2)^2 \geq 0$, so $f(w) \geq 3 = f(2)$ for every w. Equivalently, the derivative is negative below $w = 2$ and positive above it, so the function falls into that point and rises out of it. **3.** At $w = 1$ the derivative is $2(1) - 4 = -2$, so f is falling there. With $\varepsilon = 0.01$, Equation 1 predicts a change of $(0.01)(-2) = -0.02$, from

$$\begin{aligned} f(1) &= 1 - 4 + 7 \\ &= 4 \end{aligned}$$

to 3.98. The exact value is

$$\begin{aligned} f(1.01) &= (1.01)^2 - 4(1.01) + 7 \\ &= 1.0201 - 4.04 + 7 \\ &= 3.9801 \end{aligned}$$

a change of -0.0199. The prediction is off by 0.0001, which is ε^2: the linear approximation drops everything of that order and smaller, so halving the step quarters the error. That trade is the whole bargain of gradient descent, the step-by-step method of fitting models defined on the optimization page. We take a step in the direction the derivative recommends, and the recommendation is trustworthy only as far as the step is small.

Definition 0.11 (Critical point)

Definition 0.11 (Critical point)**.** Let f be a function defined on an open interval containing c. The point c is a **critical point** of f if either $f'(c) = 0$ or f is not differentiable at c (Definition 0.4).

Example 0.10 (Critical points that are and are not flat points)

Example 0.10 (Critical points that are and are not flat points)**.**

- $g(w) = |w|$ has no derivative at $w = 0$: its one-sided derivatives there are 1 and -1 (Example 0.6), so the difference quotient $\frac{|h|-0}{h}$ has no limit as $h \to 0$ (Definition 0.2). So 0 is a critical point of g but not a flat point.
- $f(w) = w^3$ has

  $$\begin{aligned} f'(0) &= 3 \cdot 0^2 \\ &= 0, \end{aligned}$$

  so 0 is a flat point of f, and hence a critical point.
- $h(w) = w^2$ is differentiable everywhere, with $h'(w) = 2w$, which is 0 only at $w = 0$. So 0 is the only critical point of h; for example, $h'(1) = 2 \neq 0$, so 1 is not one.

Theorem 0.11 (Every flat point is a critical point)

Theorem 0.11 (Every flat point is a critical point)**.** *If c is a flat point of f (Definition 0.10), then c is a critical point of f (Definition 0.11).*

Proof

Proof. A flat point c has f differentiable at c and $f'(c) = 0$. The condition "$f'(c) = 0$" is the first alternative in Definition 0.11, so c is a critical point. □

Example 0.11 (The flat points of a cubic, and a critical point that is not flat)

Example 0.11 (The flat points of a cubic, and a critical point that is not flat)**.** For $f(w) = w^3 - 3w$, the derivative is $f'(w) = 3w^2 - 3$. The code differentiates f with `D()` and evaluates f' at several points. For $g(w) = |w|$, it computes the difference quotients at 0 from both sides.

```
f_prime <- D(quote(w^3 - 3 * w), "w")
points <- c(-2, -1, 0, 1, 2)
derivs <- data.frame(w = points, f_prime = eval(f_prime, list(w = points)))
derivs
#>    w f_prime
#> 1 -2       9
#> 2 -1       0
#> 3  0      -3
#> 4  1       0
#> 5  2       9
flat <- derivs$w[derivs$f_prime == 0]

h <- c(-1e-3, 1e-3)
quotients <- abs(h) / h
quotients
#> [1] -1  1
```

The derivative is 0 at 2 of the 5 points, namely $w = -1$ and $w = 1$. These are flat points of f, so by Theorem 0.11 they are critical points. The difference quotients of g at 0 are -1 from the left and 1 from the right, so g has no derivative at 0. Thus 0 is a critical point of g (Definition 0.11), but not a flat point.

Exercise 0.15 (Maximizing a gamma-family kernel)

Exercise 0.15 (Maximizing a gamma-family kernel)**.** Adapted from Miller (2016), Question 1.1.10.
Find the maximum value of

$$f(x) = x^4 e^{-x}$$

on the interval $[0, \infty)$.

Solution 0.15

Solution 0.15. Differentiate using the product rule:

$$\begin{aligned} f'(x) &= 4x^3 e^{-x} - x^4 e^{-x} \\ &= x^3 e^{-x}(4 - x) \end{aligned}$$

On $[0, \infty)$, $e^{-x} > 0$, so $f'(x) = 0$ only at $x = 0$ and $x = 4$.
Evaluate f at the boundary and critical points:

- At $x = 0$: $f(0) = 0$.
- At $x = 4$:

$$\begin{aligned} f(4) &= 4^4 e^{-4} \\ &= 256 e^{-4} \\ &\approx 4.6888. \end{aligned}$$

- As $x \to \infty$: exponential decay dominates polynomial growth, so $\lim_{x \to \infty} x^4 e^{-x} = 0$.

Since $f(x) \geq 0$ for all $x \geq 0$ and $f(4) > 0$, the global maximum occurs at $x = 4$, with maximum value $256/e^4$.

Exercise 0.16 (Classifying critical points with the second derivative test)

Exercise 0.16 (Classifying critical points with the second derivative test)**.** Adapted from Miller (2016), Question 1.1.11.
Find the critical points of

$$f(x) = 4x^3 - 3x^2$$

and decide whether each is a local maximum, a local minimum, or an inflection point.

Solution 0.16

Solution 0.16. Compute the first derivative:

$$\begin{aligned} f'(x) &= 12x^2 - 6x \\ &= 6x(2x - 1) \end{aligned}$$

Setting $f'(x) = 0$ gives critical points at $x = 0$ and $x = 1/2$.

Compute the second derivative:

$$f''(x) = 24x - 6$$

Apply the second derivative test at each critical point:

- At $x = 0$: $f''(0) = -6 < 0$, so $x = 0$ is a strict local maximum.
- At $x = 1/2$:

$$\begin{aligned} f''(1/2) &= 24(1/2) - 6 \\ &= 6 \\ &> 0, \end{aligned}$$

  so $x = 1/2$ is a strict local minimum.

Inflection points occur where the second derivative changes sign ($f''(x) = 0$ at $x = 1/4$). Because f'' is non-zero at both critical points, neither is an inflection point.

Taylor series

Definition 0.12 (Taylor polynomial and Taylor series)

Definition 0.12 (Taylor polynomial and Taylor series)**.** Let f be a function that has at least n derivatives at a point x_0. The **Taylor polynomial** of degree n for f centered at x_0 is:

$$\begin{aligned} P_n(x) &= \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \\ &= f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n \end{aligned} \tag{2}$$

When $x_0 = 0$, $P_n(x)$ is called the **Maclaurin polynomial**. When f is infinitely differentiable and the series converges to $f(x)$ on an open interval containing x_0, the infinite sum $\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$ is the **Taylor series** of f centered at x_0.

Example 0.12 (Taylor series of the exponential function)

Example 0.12 (Taylor series of the exponential function)**.** For $f(x) = e^x$ centered at $x_0 = 0$, every derivative is

$$\begin{aligned} f^{(k)}(0) &= e^0 \\ &= 1. \end{aligned}$$

The degree-n Maclaurin polynomial is:

$$\begin{aligned} P_n(x) &= 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} \\ &= \sum_{k=0}^{n} \frac{x^k}{k!} \end{aligned}$$

Because the remainder $R_n(x) \to 0$ as $n \to \infty$ for all $x \in \mathbb{R}$, the Taylor series converges everywhere to e^x:

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

Video

Chapter 11 of the *Essence of calculus* series by 3Blue1Brown develops the geometric and polynomial intuition behind Taylor approximations:

- Taylor series shows how higher-order derivatives match the curvature and rate of change of a function near a point to build polynomial approximations (Definition 0.12).

Exercise 0.17 (Taylor polynomial of a polynomial at the origin)

Exercise 0.17 (Taylor polynomial of a polynomial at the origin)**.** Adapted from Miller (2016), Question 1.1.16.
Find the first five terms (up to degree 4) of the Taylor series for

$$f(x) = x^8 + x^4 + 3$$

at $x_0 = 0$.

Solution 0.17

Solution 0.17. Evaluate f and its first four derivatives at $x = 0$:

$$
\begin{aligned}
f(x) &= x^8 + x^4 + 3 && \implies f(0) = 3 \\
f'(x) &= 8x^7 + 4x^3 && \implies f'(0) = 0 \\
f''(x) &= 56x^6 + 12x^2 && \implies f''(0) = 0 \\
f'''(x) &= 336x^5 + 24x && \implies f'''(0) = 0 \\
f^{(4)}(x) &= 1680x^4 + 24 && \implies f^{(4)}(0) = 24
\end{aligned}
$$

By Equation 2:

$$
\begin{aligned}
P_4(x) &= 3 + 0x + \frac{0}{2!}x^2 + \frac{0}{3!}x^3 + \frac{24}{4!}x^4 \\
&= 3 + x^4
\end{aligned}
$$

Because f is already a polynomial centered at 0, its Taylor polynomial of degree 4 simply collects its terms of degree at most 4.

Exercise 0.18 (Taylor expansion of a polynomial at a nonzero point)

Exercise 0.18 (Taylor expansion of a polynomial at a nonzero point)**.** Adapted from Miller (2016), Question 1.1.17.
Find the first three terms (up to degree 2) of the Taylor series for

$$
f(x) = x^8 + x^4 + 3
$$

at $x_0 = 1$.

Solution 0.18

Solution 0.18. Evaluate f and its first two derivatives at $x_0 = 1$:

$$
\begin{aligned}
f(1) &= 1^8 + 1^4 + 3 \\
&= 5 \\
f'(1) &= 8(1)^7 + 4(1)^3 \\
&= 12 \\
f''(1) &= 56(1)^6 + 12(1)^2 \\
&= 68
\end{aligned}
$$

The degree-2 Taylor polynomial is:

$$
\begin{aligned}
P_2(x) &= f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 \\
&= 5 + 12(x-1) + 34(x-1)^2
\end{aligned}
$$

Example 0.13 (Taylor series of the cosine function)

Example 0.13 (Taylor series of the cosine function)**.** For $f(x) = \cos x$ centered at $x_0 = 0$, the derivatives follow a repeating cycle of length 4:

$$\begin{aligned} f(0) &= \cos(0) \\ &= 1, \\ f'(0) &= -\sin(0) \\ &= 0, \\ f''(0) &= -\cos(0) \\ &= -1, \\ f'''(0) &= \sin(0) \\ &= 0, \end{aligned}$$

and in general $f^{(2k)}(0) = (-1)^k$ and $f^{(2k+1)}(0) = 0$ for all $k \geq 0$. The Taylor series (Maclaurin series) converges everywhere to $\cos x$:

$$\begin{aligned} \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} \end{aligned}$$

Example 0.14 (Taylor series of the sine function)

Example 0.14 (Taylor series of the sine function)**.** For $f(x) = \sin x$ centered at $x_0 = 0$, the derivatives evaluate at 0 to $f^{(2k)}(0) = 0$ and $f^{(2k+1)}(0) = (-1)^k$ for all $k \geq 0$. The Taylor series converges everywhere to $\sin x$:

$$\begin{aligned} \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} \end{aligned}$$

Exercise 0.19 (Taylor expansion of a scaled cosine)

Exercise 0.19 (Taylor expansion of a scaled cosine)**.** Adapted from Miller (2016), Question 1.1.18.
Find the terms up to degree 4 in the Taylor series for

$$f(x) = \cos(5x)$$

at $x_0 = 0$.

Solution 0.19

Solution 0.19. Recall the standard Maclaurin series for cosine:

$$\cos u = 1 - \frac{u^2}{2!} + \frac{u^4}{4!} - \ldots$$

Substitute $u = 5x$:

$$\begin{aligned}\cos(5x) &= 1 - \frac{(5x)^2}{2!} + \frac{(5x)^4}{4!} - \ldots \\ &= 1 - \frac{25}{2}x^2 + \frac{625}{24}x^4 - \ldots\end{aligned}$$

Remark: Substituting into known Taylor expansions avoids computing high-order derivatives by hand.

Exercise 0.20 (Taylor series of a power of a function)

Exercise 0.20 (Taylor series of a power of a function)**.** Adapted from Miller (2016), Question 1.1.19.
Find the terms up to degree 4 in the Taylor series for

$$f(x) = \cos^3(5x)$$

at $x_0 = 0$.

Solution 0.20

Solution 0.20. From Exercise 0.19, the expansion of $\cos(5x)$ up to order x^4 is:

$$\cos(5x) = 1 - \frac{25}{2}x^2 + \frac{625}{24}x^4 + O(x^6)$$

Cube this expansion using $(1-y)^3 = 1 - 3y + 3y^2 - y^3$ with $y = \frac{25}{2}x^2 - \frac{625}{24}x^4$:

$$\begin{aligned}\cos^3(5x) &= 1 - 3\left(\frac{25}{2}x^2 - \frac{625}{24}x^4\right) + 3\left(\frac{25}{2}x^2\right)^2 + O(x^6) \\ &= 1 - \frac{75}{2}x^2 + \frac{625}{8}x^4 + \frac{1875}{4}x^4 + O(x^6) \\ &= 1 - \frac{75}{2}x^2 + \frac{4375}{8}x^4 + O(x^6)\end{aligned}$$

Exercise 0.21 (Linear Taylor polynomial of the exponential)

Exercise 0.21 (Linear Taylor polynomial of the exponential)**.** Adapted from Miller (2016), Question 1.1.20.
Find the first two terms (degree 1) of the Taylor series for

$$f(x) = e^x$$

at $x_0 = 0$.

Solution 0.21

Solution 0.21. Since

$$\begin{aligned} f(0) &= e^0 \\ &= 1 \end{aligned}$$

and

$$\begin{aligned} f'(0) &= e^0 \\ &= 1 : \end{aligned}$$

$$\begin{aligned} P_1(x) &= f(0) + f'(0)x \\ &= 1 + x \end{aligned}$$

The full Taylor series is

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \\ &= 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots . \end{aligned}$$

Exercise 0.22 (Taylor series of a composite exponential)

Exercise 0.22 (Taylor series of a composite exponential)**.** Adapted from Miller (2016), Question 1.1.21.
Find the first six terms (up to degree 5) of the Taylor series for

$$f(x) = \exp\{x^8\}$$

at $x_0 = 0$.

Solution 0.22

Solution 0.22. Substitute $u = x^8$ into $e^u = 1 + u + \frac{u^2}{2!} + ...$:

$$\exp\{x^8\} = 1 + x^8 + \frac{x^{16}}{2} + ...$$

Because the lowest-order variable term is x^8, all coefficients for powers x^1, x^2, x^3, x^4, x^5 are zero:

$$\begin{aligned} P_5(x) &= 1 + 0x + 0x^2 + 0x^3 + 0x^4 + 0x^5 \\ &= 1 \end{aligned}$$

Exercise 0.23 (Taylor expansion of the normal density)

Exercise 0.23 (Taylor expansion of the normal density)**.** Adapted from Miller (2016), Question 1.1.22.
Find the terms up to degree 3 of the Taylor series for the standard normal density

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}$$

at $x_0 = 0$.

Solution 0.23

Solution 0.23. Using $e^u = 1 + u + O(u^2)$ with $u = -x^2/2$:

$$\exp\left\{-\frac{x^2}{2}\right\} = 1 - \frac{x^2}{2} + O(x^4)$$

Multiplying by the normalization constant $1/\sqrt{2\pi}$:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} - \frac{x^2}{2\sqrt{2\pi}} + O(x^4)$$

Up to degree 3, the odd powers vanish by symmetry, so:

$$P_3(x) = \frac{1}{\sqrt{2\pi}} - \frac{x^2}{2\sqrt{2\pi}}$$

Exercise 0.24 (Taylor expansion of the square root)

Exercise 0.24 (Taylor expansion of the square root)**.** Adapted from Miller (2016),

Question 1.1.23.
Find the first three terms (up to degree 2) of the Taylor series for

$$f(x) = \sqrt{x}$$

at $x_0 = 1/3$.

Solution 0.24

Solution 0.24. Compute the values at $x_0 = 1/3$:

$$\begin{aligned} f(1/3) &= \left(\frac{1}{3}\right)^{1/2} \\ &= \frac{1}{\sqrt{3}} \\ &= \frac{\sqrt{3}}{3} \\ f'(x) &= \frac{1}{2}x^{-1/2} \\ \implies f'(1/3) &= \frac{1}{2}\sqrt{3} \\ &= \frac{\sqrt{3}}{2} \\ f''(x) &= -\frac{1}{4}x^{-3/2} \\ \implies f''(1/3) &= -\frac{1}{4}\cdot 3\sqrt{3} \\ &= -\frac{3\sqrt{3}}{4} \end{aligned}$$

The degree-2 Taylor polynomial is:

$$P_2(x) = \frac{\sqrt{3}}{3} + \frac{\sqrt{3}}{2}\left(x - \frac{1}{3}\right) - \frac{3\sqrt{3}}{8}\left(x - \frac{1}{3}\right)^2$$

Exercise 0.25 (Taylor expansion of a fractional power)

Exercise 0.25 (Taylor expansion of a fractional power)**.** Adapted from Miller (2016), Question 1.1.24.
Find the first three terms of the Taylor series for

$$f(x) = (1+x)^{1/3}$$

at $x_0 = 1/2$.

Solution 0.25

Solution 0.25. At $x_0 = 1/2$, $1 + x_0 = 3/2$:

$$\begin{aligned} f(1/2) &= \left(\frac{3}{2}\right)^{1/3} \\ f'(x) &= \frac{1}{3}(1+x)^{-2/3} \\ \implies f'(1/2) &= \frac{1}{3}\left(\frac{3}{2}\right)^{-2/3} \\ &= \frac{1}{3}\left(\frac{2}{3}\right)^{2/3} \\ f''(x) &= -\frac{2}{9}(1+x)^{-5/3} \\ \implies f''(1/2) &= -\frac{2}{9}\left(\frac{3}{2}\right)^{-5/3} \\ &= -\frac{2}{9}\left(\frac{2}{3}\right)^{5/3} \end{aligned}$$

Dividing $f''(1/2)$ by 2! gives $-\frac{1}{9}\left(\frac{2}{3}\right)^{5/3}$. The Taylor polynomial is:

$$P_2(x) = \left(\frac{3}{2}\right)^{1/3} + \frac{1}{3}\left(\frac{2}{3}\right)^{2/3}\left(x - \frac{1}{2}\right) - \frac{1}{9}\left(\frac{2}{3}\right)^{5/3}\left(x - \frac{1}{2}\right)^2$$

Exercise 0.26 (Taylor expansion of $x \log x$)

Exercise 0.26 (Taylor expansion of $x \log x$)**.** Adapted from Miller (2016), Question 1.1.25.
Find the first three terms (up to degree 2) of the Taylor series for

$$f(x) = x \log x$$

at $x_0 = 1$.

Solution 0.26

Solution 0.26. Evaluate derivatives at $x_0 = 1$:

$$
\begin{aligned}
f(1) &= 1 \log 1 \\
&= 0 \\
f'(x) &= \log x + 1 \\
\implies f'(1) &= 0 + 1 \\
&= 1 \\
f''(x) &= \frac{1}{x} \implies f''(1) = 1
\end{aligned}
$$

Therefore:

$$
\begin{aligned}
P_2(x) &= f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 \\
&= (x-1) + \frac{1}{2}(x-1)^2
\end{aligned}
$$

Exercise 0.27 (Maclaurin series of $\log(1+x)$)

Exercise 0.27 (Maclaurin series of $\log(1+x)$)**.** Adapted from Miller (2016), Question 1.1.26.
Find the first three terms of the Taylor series for

$$f(x) = \log(1+x)$$

at $x_0 = 0$.

Solution 0.27

Solution 0.27. Differentiating repeatedly at $x_0 = 0$:

$$
\begin{aligned}
f(0) &= \log 1 \\
&= 0 \\
f'(x) &= (1+x)^{-1} \implies f'(0) = 1 \\
f''(x) &= -(1+x)^{-2} \implies f''(0) = -1 \\
f'''(x) &= 2(1+x)^{-3} \implies f'''(0) = 2
\end{aligned}
$$

The Taylor polynomial of degree 3 is:

$$P_3(x) = x - \frac{x^2}{2} + \frac{x^3}{3}$$

Exercise 0.28 (Maclaurin series of $\log(1-x)$)

Exercise 0.28 (Maclaurin series of $\log(1-x)$)**.** Adapted from Miller (2016), Question 1.1.27.
Find the terms up to degree 3 of the Taylor series for

$$f(x) = \log(1-x)$$

at $x_0 = 0$.

Solution 0.28

Solution 0.28. Substitute $u = -x$ into the expansion for $\log(1+u)$ from Exercise 0.27:

$$\begin{aligned}\log(1-x) &= (-x) - \frac{(-x)^2}{2} + \frac{(-x)^3}{3} - \ldots \\ &= -x - \frac{x^2}{2} - \frac{x^3}{3} - \ldots\end{aligned}$$

Exercise 0.29 (Taylor series of $\log((1-x)e^x)$)

Exercise 0.29 (Taylor series of $\log((1-x)e^x)$)**.** Adapted from Miller (2016), Question 1.1.28.
Find the first two non-zero terms of the Taylor series for

$$f(x) = \log((1-x)e^x)$$

at $x_0 = 0$.

Solution 0.29

Solution 0.29. Expand using logarithm properties:

$$\begin{aligned}f(x) &= \log(1-x) + \log(e^x) \\ &= \log(1-x) + x\end{aligned}$$

Substitute the series for $\log(1-x)$ from Exercise 0.28:

$$\begin{aligned}f(x) &= \left(-x - \frac{x^2}{2} - \frac{x^3}{3} - \ldots\right) + x \\ &= -\frac{x^2}{2} - \frac{x^3}{3} - \ldots\end{aligned}$$

The constant and linear terms vanish ($f(0) = 0$, $f'(0) = 0$). The first two non-zero terms

are:

$$-\frac{x^2}{2} - \frac{x^3}{3}$$

Exercise 0.30 (Taylor series of a product of known expansions)

Exercise 0.30 (Taylor series of a product of known expansions)**.** Adapted from Miller (2016), Question 1.1.29.
Find the terms up to degree 2 of the Taylor series for

$$f(x) = \cos(x)\log(1+x)$$

at $x_0 = 0$.

Solution 0.30

Solution 0.30. Multiply the known Maclaurin expansions:

$$\cos x = 1 - \frac{x^2}{2} + O(x^4)$$
$$\log(1+x) = x - \frac{x^2}{2} + O(x^3)$$

Multiplying and dropping terms of order x^3 and higher:

$$\begin{aligned} f(x) &= \left(1 - \frac{x^2}{2} + \ldots\right)\left(x - \frac{x^2}{2} + \ldots\right) \\ &= x - \frac{x^2}{2} + O(x^3) \end{aligned}$$

Exercise 0.31 (Taylor series of $\log(1+2x)$)

Exercise 0.31 (Taylor series of $\log(1+2x)$)**.** Adapted from Miller (2016), Question 1.1.30.
Find the first two non-zero terms of the Taylor series for

$$f(x) = \log(1+2x)$$

at $x_0 = 0$.

Solution 0.31

Solution 0.31. Substitute $u = 2x$ into the Maclaurin series $\log(1+u) = u - \frac{u^2}{2} + \ldots$:

$$\begin{aligned}\log(1+2x) &= (2x) - \frac{(2x)^2}{2} + \ldots \\ &= 2x - 2x^2 + \ldots\end{aligned}$$

Integration

Integration is the inverse operation of differentiation: it recovers a function from its derivative and accumulates quantities such as areas, totals, and probabilities. We begin with antiderivatives, then state basic integration rules, and conclude with the Fundamental Theorem of Calculus and a worked example from probability.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist introduces integral calculus:

- Intro to Integral Calculus
- What Integral Calculus Is

Antiderivatives

Definition 0.13 (Antiderivative)

Definition 0.13 (Antiderivative)**.** A function F is an **antiderivative** of f on an interval I if:

$$\frac{\partial}{\partial x} F(x) = f(x), \quad \forall x \in I$$

Finding an antiderivative of f is called **antidifferentiation**, or **antidifferentiating** f. (Larson and Edwards 2018, sec. 4.1, pp. 248–249)

Definition 0.14 (Indefinite integral)

Definition 0.14 (Indefinite integral)**.** The **indefinite integral** of f is the family of all antiderivatives (Definition 0.13) of f:

$$\int f(x)\,dx = F(x) + C$$

where F is any one antiderivative of f and C is an arbitrary constant of integration. (Larson and Edwards 2018, sec. 4.1, pp. 248–249)

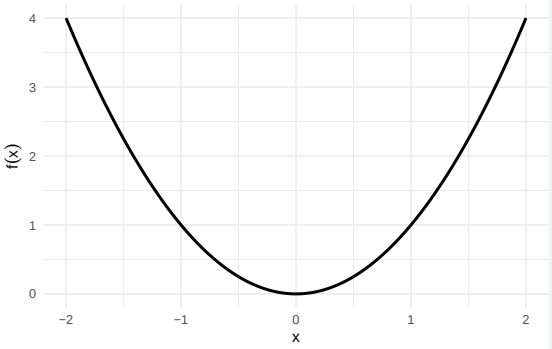
Example 0.15 (Antiderivative of x^2)

Example 0.15 (Antiderivative of x^2)**.** For $f(x) = x^2$, an antiderivative is $F(x) = \frac{x^3}{3}$, since

$$\begin{aligned}\frac{\partial}{\partial x}\frac{x^3}{3} &= x^2 \\ &= f(x).\end{aligned}$$

Adding any constant C gives another antiderivative; for example, with $C = 7$, $F(x) = \frac{x^3}{3} + 7$ also satisfies $F'(x) = x^2$, since adding a constant does not change the derivative. $G(x) = x^3$ is not an antiderivative of x^2: $G'(x) = 3x^2 \neq x^2$ for $x \neq 0$.
See Figure 2.

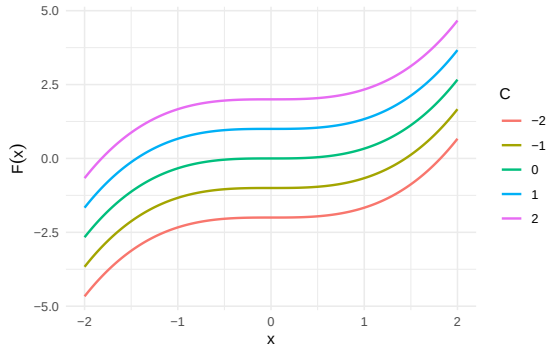
```
ggplot2::ggplot() +
  ggplot2::geom_function(fun = \(x) x^2) +
  ggplot2::labs(x = "x", y = expression(f(x))) +
  ggplot2::theme_minimal()
```



(a) The function $f(x) = x^2$.

```
x_seq <- seq(x_lim[1], x_lim[2], length.out = 200)
df <- C_vals |>
  lapply(\(C) {
    tibble::tibble(
      x = x_seq,
      y = x_seq^3 / 3 + C,
      C = factor(C)
    )
  }) |>
  dplyr::bind_rows()

ggplot2::ggplot(df, ggplot2::aes(x = x, y = y, color = C)) +
  ggplot2::geom_line(linewidth = 0.8) +
  ggplot2::labs(x = "x", y = expression(F(x)), color = "C") +
  ggplot2::theme_minimal()
```



(b) Family of antiderivatives $F(x) = x^3/3 + C$.

Figure 2: The function $f(x) = x^2$ and five antiderivatives $F(x) = x^3/3 + C$ for $C \in \{-2, -1, 0, 1, 2\}$. Each antiderivative has the same derivative f; they differ only by a vertical shift.

Theorem 0.12 (Basic integration rules)

Theorem 0.12 (Basic integration rules)**.** *Each antiderivative in the table is defined only up to an arbitrary constant C (see Definition 0.13); the table omits $+C$ from every row for brevity.*

Function $f(x)$	*Antiderivative $F(x)$*	*Condition*
c	cx	—

x^n	$\dfrac{x^{n+1}}{n+1}$	$n \neq -1$
$\dfrac{1}{x}$	$\log\{\lvert x\rvert\}$	$x \neq 0$
e^x	e^x	*(self-antiderivative)*
e^{cx}	$\dfrac{1}{c}e^{cx}$	$c \neq 0$
$\sin x$	$-\cos x$	—
$\cos x$	$\sin x$	—
$c \cdot f(x)$	$c \cdot F(x)$	—
$f(x) + g(x)$	$F(x) + G(x)$	—

The first two rows, the trigonometric rows ($\sin x$, $\cos x$), and the bottom two rows (linearity) are from (Larson and Edwards 2018, sec. 4.1, p. 250 "Basic Integration Rules"); $1/x$ is from (Larson and Edwards 2018, sec. 5.2, Theorem 5.5, p. 324); e^x and e^{cx} are from (Larson and Edwards 2018, sec. 5.4, Theorem 5.12, p. 346).

Example 0.16 (Antiderivative of $3x^2 - 1$)

Example 0.16 (Antiderivative of $3x^2 - 1$)**.** By the power rule ($n = 2$) and linearity from Theorem 0.12:

$$\begin{aligned}\int (3x^2 - 1)\, dx &= 3 \cdot \frac{x^3}{3} - x + C \\ &= x^3 - x + C.\end{aligned}$$

Verify by differentiating:

$$\begin{aligned}\frac{\partial}{\partial x}(x^3 - x + C) &= 3x^2 - 1 \\ &= f(x),\end{aligned}$$

as required.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on these rules:

- The Integral Calculus Rules
- Indefinite Integral Exercises

Regularity Conditions

Definition 0.15 (Differentiable on an interval)

Definition 0.15 (Differentiable on an interval)**.** A function f is **differentiable on** an interval if it is differentiable (Definition 0.4) at every point of the interval other than its endpoints, and, at each endpoint that belongs to the interval, it has the one-sided derivative (Definition 0.6) from inside the interval: the right-hand derivative at the left endpoint and the left-hand derivative at the right endpoint.
(Larson and Edwards 2018, sec. 2.1, p. 100)

Example 0.17 (Differentiable on one interval but not another)

Example 0.17 (Differentiable on one interval but not another)**.**

- $f(x) = x^2$ is differentiable on $[0, 1]$: the computation of Example 0.4 with 3 replaced by any c gives $\frac{f(c+h)-f(c)}{h} = 2c + h$, which tends to $2c$, including the one-sided limits at the endpoints 0 and 1.
- $g(x) = \sqrt[3]{x}$ is differentiable on $[1, 2]$, but not on $[-1, 1]$: the point 0 is in $[-1, 1]$ and is not an endpoint, and Example 0.4 shows g is not differentiable there.

Definition 0.16 (Continuous function)

Definition 0.16 (Continuous function)**.** A function f is **continuous at** $x = c$ if all three conditions hold:

1. $f(c)$ is defined,
2. $\lim_{x\to c} f(x)$ exists, and
3. $\lim_{x\to c} f(x) = f(c)$.

If any of the three conditions fails, f is **discontinuous** at c.
(Larson and Edwards 2018, sec. 1.4, p. 73)

Example 0.18 (A continuous function, and one failure of each condition)

Example 0.18 (A continuous function, and one failure of each condition)**.**

- $f(x) = x^2$ is continuous at $c = 1$: $f(1) = 1$ is defined, and

$$\begin{aligned}\lim_{x\to 1} x^2 &= 1 \\ &= f(1).\end{aligned}$$

- $g(x) = \frac{x^2-1}{x-1}$ fails condition 1 at $c = 1$: $g(1)$ is not defined (it would divide by zero), even though

$$\begin{aligned}\lim_{x\to 1} g(x) &= \lim_{x\to 1}(x+1)\\ &= 2\end{aligned}$$

  exists ($x^2 - 1 = (x-1)(x+1)$, and the factor $x - 1$ cancels for $x \neq 1$).

- The step function $H(x) = 1$ for $x \geq 0$ and $H(x) = 0$ for $x < 0$ fails condition 2 at $c = 0$: values to the left are all 0 and values to the right are all 1, so $\lim_{x\to 0} H(x)$ does not exist.

- $k(x) = x^2$ for $x \neq 1$, with $k(1) = 5$, fails condition 3 at $c = 1$: $\lim_{x\to 1} k(x) = 1$ exists but differs from $k(1) = 5$.

Definition 0.17 (Jump discontinuity)

Definition 0.17 (Jump discontinuity)**.** A function f has a **jump discontinuity** at c if both one-sided limits (Definition 0.2) $\lim_{x\to c^-} f(x)$ and $\lim_{x\to c^+} f(x)$ exist but are not equal. Then $\lim_{x\to c} f(x)$ does not exist, so f is discontinuous at c (Definition 0.16).

Example 0.19 (A jump discontinuity, and a discontinuity that is not a jump)

Example 0.19 (A jump discontinuity, and a discontinuity that is not a jump)**.**

- The step function H of Example 0.2 has a jump discontinuity at 0: $\lim_{x\to 0^-} H(x) = 0$ and $\lim_{x\to 0^+} H(x) = 1$ both exist, and $0 \neq 1$. The size of the jump is $1 - 0 = 1$.
- $g(x) = \frac{x^2-1}{x-1}$ of Example 0.18 is discontinuous at 1, because $g(1)$ is not defined, but it does not have a jump discontinuity there: for $x \neq 1$, $g(x) = x + 1$, so both one-sided limits at 1 equal $1 + 1 = 2$.

Definition 0.18 (Continuous on a closed interval)

Definition 0.18 (Continuous on a closed interval)**.** A function f is **continuous on** a closed interval $[a, b]$ if all three conditions hold:

1. f is continuous (Definition 0.16) at every point of the open interval (a, b),
2. $\lim_{x\to a^+} f(x) = f(a)$, and
3. $\lim_{x\to b^-} f(x) = f(b)$.

(Larson and Edwards 2018, sec. 1.4, p. 73)

Example 0.20 (Continuity on $[0,1]$ uses one-sided limits at the endpoints)

Example 0.20 (Continuity on $[0,1]$ uses one-sided limits at the endpoints)**.** Let $f(x) = \sqrt{x}$, the square root, defined for $x \geq 0$. Because f is undefined for $x < 0$, only the right-hand limit of f at 0 makes sense, and Definition 0.18 asks only for that one-sided limit at the endpoint 0. Here f is continuous at every point of $(0,1)$,

$$\begin{aligned} \lim_{x \to 0^+} \sqrt{x} &= 0 \\ &= f(0), \end{aligned}$$

and

$$\begin{aligned} \lim_{x \to 1^-} \sqrt{x} &= 1 \\ &= f(1), \end{aligned}$$

so f is continuous on $[0,1]$ (Definition 0.18).

Example 0.21 (A function not continuous on $[0,1]$)

Example 0.21 (A function not continuous on $[0,1]$)**.** Let $f(x) = 0$ for $0 \leq x < 1$ and $f(1) = 2$. Conditions 1 and 2 of Definition 0.18 hold, but

$$\begin{aligned} \lim_{x \to 1^-} f(x) &= 0 \\ &\neq 2 \\ &= f(1), \end{aligned}$$

so condition 3 fails and f is not continuous on $[0,1]$.

Definition 0.19 (Partition of an interval)

Definition 0.19 (Partition of an interval)**.** A **partition** $\mathcal{P}$ of a closed interval $[a,b]$ is a finite list of points

$$\begin{aligned} a &= x_0 \\ &< x_1 \\ &< \cdots \\ &< x_n \\ &= b. \end{aligned}$$

It splits $[a,b]$ into the n subintervals $[x_{i-1}, x_i]$, of widths $\Delta x_i \stackrel{\text{def}}{=} x_i - x_{i-1}$.
(Larson and Edwards 2018, sec. 4.3)

Example 0.22 (A partition of $[0,1]$)

Example 0.22 (A partition of $[0,1]$)**.** The points $0 < 0.25 < 0.5 < 1$ form a partition of $[0,1]$ with $n = 3$ subintervals, of widths $\Delta x_1 = 0.25$, $\Delta x_2 = 0.25$, and $\Delta x_3 = 0.5$.

Example 0.23 (Lists that are not partitions of $[0,1]$)

Example 0.23 (Lists that are not partitions of $[0,1]$)**.**

- $0, 0.5, 0.25, 1$ is not a partition: the points are not increasing, since $0.5 > 0.25$.
- $0 < 0.5$ is not a partition of $[0,1]$: its last point is 0.5, not $b = 1$.

Definition 0.20 (Mesh of a partition)

Definition 0.20 (Mesh of a partition)**.** The **mesh** of a partition $\mathcal{P}$ (Definition 0.19) is its largest subinterval width,

$$\|\mathcal{P}\| \stackrel{\text{def}}{=} \max_{i \in \{1,\ldots,n\}} \Delta x_i,$$

where n is the number of subintervals of $\mathcal{P}$.
(Larson and Edwards 2018, sec. 4.3)

Example 0.24 (The mesh of a partition of $[0,1]$)

Example 0.24 (The mesh of a partition of $[0,1]$)**.** For the partition of $[0,1]$ in Example 0.22, with widths $\Delta x_1 = 0.25$, $\Delta x_2 = 0.25$, and $\Delta x_3 = 0.5$, the mesh is the largest of these widths, $\|\mathcal{P}\| = 0.5$.

Definition 0.21 (Riemann sum)

Definition 0.21 (Riemann sum)**.** Let f be a function on $[a,b]$, let $\mathcal{P}$ be a partition

$$\begin{aligned} a &= x_0 \\ &< x_1 \\ &< \cdots \\ &< x_n \\ &= b \end{aligned}$$

of $[a,b]$ (Definition 0.19), and choose a **sample point** x_i^* in each subinterval $[x_{i-1}, x_i]$. The **Riemann sum** of f for $\mathcal{P}$ and these sample points is

$$\sum_{i=1}^{n} f(x_i^*)\,\Delta x_i,$$

where $\Delta x_i = x_i - x_{i-1}$.
(Larson and Edwards 2018, sec. 4.3)

Example 0.25 (A Riemann sum for x^2 on $[0,1]$)

Example 0.25 (A Riemann sum for x^2 on $[0,1]$)**.** Let $f(x) = x^2$, take the partition $0 < 0.25 < 0.5 < 1$ of Example 0.22, with widths $\Delta x_1 = 0.25$, $\Delta x_2 = 0.25$, and $\Delta x_3 = 0.5$, and take each sample point at the right end of its subinterval: $x_1^* = 0.25$, $x_2^* = 0.5$, and $x_3^* = 1$. Then

$$\begin{aligned}
\sum_{i=1}^{3} f(x_i^*)\,\Delta x_i &= f(0.25)\cdot 0.25 + f(0.5)\cdot 0.25 + f(1)\cdot 0.5 && \text{(write out the three terms)}\\
&= 0.0625\cdot 0.25 + 0.25\cdot 0.25 + 1\cdot 0.5 && \text{(evaluate } f(x)=x^2\text{)}\\
&= 0.015625 + 0.0625 + 0.5 && \text{(multiply)}\\
&= 0.578125 && \text{(add)}
\end{aligned}$$

Definition 0.22 (Riemann integral (definite integral))

Definition 0.22 (Riemann integral (definite integral))**.** Let f be a bounded function on $[a,b]$. For each partition $\mathcal{P}$ of $[a,b]$ (Definition 0.19), choose a sample point x_i^* in each subinterval $[x_{i-1}, x_i]$. The **Riemann integral** of f over $[a,b]$ (also called the **definite integral** of f from a to b) is the limit of the Riemann sums (Definition 0.21) as the mesh (Definition 0.20) shrinks to zero:

$$\int_a^b f(x)\,dx \overset{\text{def}}{=} \lim_{\|\mathcal{P}\|\to 0} \sum_{i=1}^{n} f(x_i^*)\,\Delta x_i,$$

when that limit exists and has the same value for every choice of the partitions and of the sample points x_i^*.
(Larson and Edwards 2018, sec. 4.3, p. 272)

Definition 0.23 (Riemann integrable)

Definition 0.23 (Riemann integrable)**.** A bounded function f is **Riemann integrable on** $[a,b]$ if the Riemann sums (Definition 0.21) $\sum_{i=1}^{n} f(x_i^*)\,\Delta x_i$, over partitions $\mathcal{P}$ of $[a,b]$ (Definition 0.19) with a sample point x_i^* in each subinterval $[x_{i-1}, x_i]$, approach a real-number limit as the mesh $\|\mathcal{P}\|$ (Definition 0.20) shrinks to zero, and that limit is the

same for every choice of the partitions and of the sample points.
(Larson and Edwards 2018, sec. 4.3, p. 272)

Example 0.26 (A constant function is integrable)

Example 0.26 (A constant function is integrable)**.** Let $f(x) = 2$ on $[0, 3]$. For every partition and every choice of sample points,

$$\begin{aligned}
\sum_{i=1}^{n} f(x_i^*)\,\Delta x_i &= \sum_{i=1}^{n} 2\,\Delta x_i && (f \text{ is 2 everywhere}) \\
&= 2\sum_{i=1}^{n} \Delta x_i && (\text{factor out the constant}) \\
&= 2 \cdot (3 - 0) && (\text{the widths add up to the length of } [0, 3]) \\
&= 6, && (\text{multiply})
\end{aligned}$$

so the sums have the same limit, 6, for every choice: f is Riemann integrable on $[0, 3]$ (Definition 0.23), and $\int_0^3 2\,dx = 6$ (Definition 0.22).

Definition 0.24 (Integrand)

Definition 0.24 (Integrand)**.** In an integral such as $\int_a^b f(x)\,dx$ (Definition 0.22) or $\int f(x)\,dx$ (Definition 0.14), the function f being integrated is the **integrand**.

Example 0.27 (Integrands)

Example 0.27 (Integrands)**.**

- In $\int_0^3 2\,dx = 6$ (Example 0.26), the integrand is the constant function $f(x) = 2$, and the limits of integration are 0 and 3.
- In $\int (3x^2 - 1)\,dx = x^3 - x + C$ (Example 0.16), the integrand is $f(x) = 3x^2 - 1$; its value at $x = 2$ is $3 \cdot 4 - 1 = 11$.

Remark 0.2 (Riemann integrable functions and Riemann integrals)

Remark 0.2 (Riemann integrable functions and Riemann integrals). A bounded function f is Riemann integrable on $[a, b]$ exactly when its Riemann integral $\int_a^b f(x)\,dx$ (Definition 0.22) exists; the integral is the common limit of the sums. For example, let $g(x) = 1$ when x is rational and $g(x) = 0$ when x is irrational, on $[0, 1]$. Every subinterval contains both rational and irrational points. Choosing every sample point rational gives $\sum_{i=1}^{n} 1 \cdot \Delta x_i = 1$ for every partition, because the widths Δx_i add up to the length $1 - 0 = 1$

of $[0,1]$, and choosing every sample point irrational gives $\sum_{i=1}^{n} 0 \cdot \Delta x_i = 0$. The two limits differ, so g is not Riemann integrable on $[0,1]$, and $\int_0^1 g(x)\,dx$ does not exist.

Definition 0.25 (Equal-width Riemann sum)

Definition 0.25 (Equal-width Riemann sum)**.** For a bounded function f on $[a,b]$ and a positive integer n, split $[a,b]$ into n subintervals of equal width $\Delta x \stackrel{\text{def}}{=} (b-a)/n$, and let x_i^* be any point in the i-th subinterval. The **equal-width Riemann sum** is

$$S_n \stackrel{\text{def}}{=} \sum_{i=1}^{n} f(x_i^*)\,\Delta x.$$

Example 0.28 (An equal-width Riemann sum)

Example 0.28 (An equal-width Riemann sum)**.** Let $f(x) = x^2$ on $[0,1]$, with $n = 2$, so $\Delta x = 1/2$, and take each sample point at the right end of its subinterval: $x_1^* = \frac{1}{2}$ and $x_2^* = 1$. Then

$$\begin{aligned} S_2 &= f\left(\tfrac{1}{2}\right) \cdot \tfrac{1}{2} + f(1) \cdot \tfrac{1}{2} && \text{(equal-width Riemann sum with } n = 2) \\ &= \tfrac{1}{4} \cdot \tfrac{1}{2} + 1 \cdot \tfrac{1}{2} && \text{(evaluate } f(x) = x^2) \\ &= \tfrac{5}{8} && \text{(add)} \end{aligned}$$

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on approximating an area by a sum of simple pieces, as a Riemann sum does, and on computing such sums in Python:

- The Method of Exhaustion
- Numeric Integration with Python

Before stating the Fundamental Theorem of Calculus, we record two prerequisite results. The usual statement of the Fundamental Theorem of Calculus assumes that the integrand f is continuous on $[a,b]$; continuity is sufficient there, though not necessary. The two results are "differentiability implies continuity", which says where continuity comes from, and "continuity implies integrability", which says what continuity buys us.

Theorem 0.13 (Differentiability implies continuity)

Theorem 0.13 (Differentiability implies continuity)**.** *If f is differentiable at $x = c$, then*

f is continuous at $x = c$.
(Larson and Edwards 2018, Theorem 2.1, p. 106)

Proof

Proof. Because $f'(c)$ exists, $f(c)$ is defined, and:

$$\begin{aligned}
\lim_{h\to 0}(f(c+h)-f(c)) &= \lim_{h\to 0}\left(\frac{f(c+h)-f(c)}{h}\cdot h\right) && \text{(multiply and divide by } h\neq 0\text{)}\\
&= \left(\lim_{h\to 0}\frac{f(c+h)-f(c)}{h}\right)\cdot\left(\lim_{h\to 0} h\right) && \text{(limit of a product, both limits exist)}\\
&= f'(c)\cdot 0 && \text{(definition of } f'(c)\text{)}\\
&= 0 && \text{(multiply)}
\end{aligned}$$

So $\lim_{h\to 0} f(c+h) = f(c)$, which is $\lim_{x\to c} f(x) = f(c)$ with $x = c+h$; all three conditions of Definition 0.16 hold. □

Example 0.29 (Differentiable, hence continuous: $x^3 - x$)

Example 0.29 (Differentiable, hence continuous: $x^3 - x$)**.** $f(x) = x^3 - x$ is differentiable everywhere (with derivative $f'(x) = 3x^2 - 1$), so by Theorem 0.13 it is continuous everywhere.

Example 0.30 (Continuous but not differentiable: $|x|$)

Example 0.30 (Continuous but not differentiable: $|x|$)**.** The absolute-value function $f(x) = |x|$ is continuous at $x = 0$, since

$$\begin{aligned}
\lim_{x\to 0}|x| &= 0\\
&= |0|,
\end{aligned}$$

but it is not differentiable at $x = 0$: its left-hand derivative there is -1 and its right-hand derivative is $+1$ (Example 0.6).
This counterexample shows that the converse of Theorem 0.13 fails: continuity does not imply differentiability. See Figure 3.

```r
ggplot2::ggplot() +
  ggplot2::geom_function(fun = abs, xlim = c(-2, 2), linewidth = 1) +
  ggplot2::geom_point(ggplot2::aes(x = 0, y = 0), size = 3) +
  ggplot2::labs(x = "x", y = expression(f(x) == abs(x))) +
  ggplot2::theme_minimal()
```

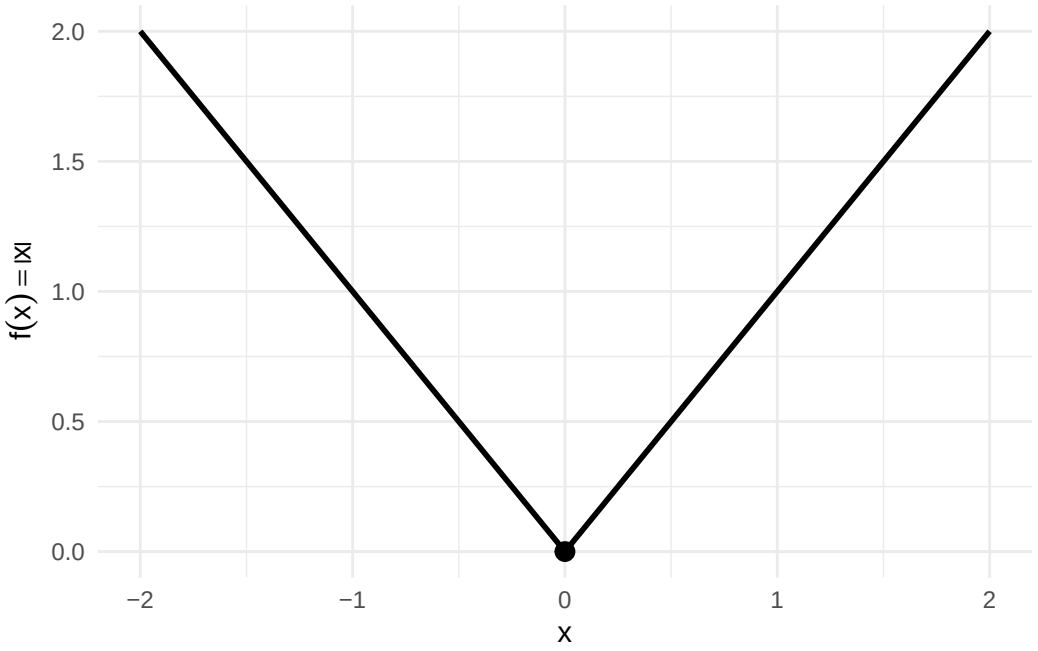


Figure 3: $f(x) = |x|$ has a sharp corner at $x = 0$ (not differentiable there) but is continuous everywhere: no gaps or jumps.

Theorem 0.14 (Continuity implies integrability)

Theorem 0.14 (Continuity implies integrability)**.** *If f is continuous on the closed interval $[a, b]$, then f is integrable on $[a, b]$ (i.e., the Riemann integral $\int_a^b f(x)\,dx$ exists and is finite).*
(Larson and Edwards 2018, Theorem 4.4, p. 272)

Example 0.31 (Continuous, hence integrable: polynomials)

Example 0.31 (Continuous, hence integrable: polynomials)**.** Every polynomial is continuous on $\mathbb{R}$, so by Theorem 0.14 every polynomial is integrable on every closed interval $[a, b]$.

Example 0.32 (Integrable but not continuous: a step function)

Example 0.32 (Integrable but not continuous: a step function)**.** Let $f(x) = 0$ for $x < \frac{1}{2}$ and $f(x) = 1$ for $x \geq \frac{1}{2}$. Then f has a jump discontinuity (Definition 0.17) at $x = \frac{1}{2}$, but it is integrable on $[0, 1]$:

$$\begin{aligned}\int_0^1 f(x)\,dx &= \int_0^{1/2} 0\,dx + \int_{1/2}^1 1\,dx \\ &= 0 + \tfrac{1}{2} \\ &= \tfrac{1}{2}.\end{aligned}$$

This counterexample shows that the converse of Theorem 0.14 fails: integrability does not imply continuity. See Figure 4.

```r
step_df <- tibble::tibble(
  x = c(0, 0.5, 0.5, 1),
  y = c(0, 0, 1, 1),
  segment = c("left", "left", "right", "right")
)

ggplot2::ggplot() +
  ggplot2::geom_rect(
    ggplot2::aes(xmin = 0.5, xmax = 1, ymin = 0, ymax = 1),
    fill = "steelblue", alpha = 0.3
  ) +
  ggplot2::geom_line(
    data = step_df,
    ggplot2::aes(x = x, y = y, group = segment),
    linewidth = 1
  ) +
  ggplot2::geom_point(ggplot2::aes(x = 0.5, y = 0), shape = 1, size = 3) +
  ggplot2::geom_point(ggplot2::aes(x = 0.5, y = 1), shape = 16, size = 3) +
  ggplot2::scale_x_continuous(
    breaks = c(0, 0.5, 1),
    labels = c("0", "1/2", "1")
  ) +
  ggplot2::scale_y_continuous(limits = c(-0.1, 1.2)) +
  ggplot2::labs(x = "x", y = "f(x)") +
  ggplot2::theme_minimal()
```

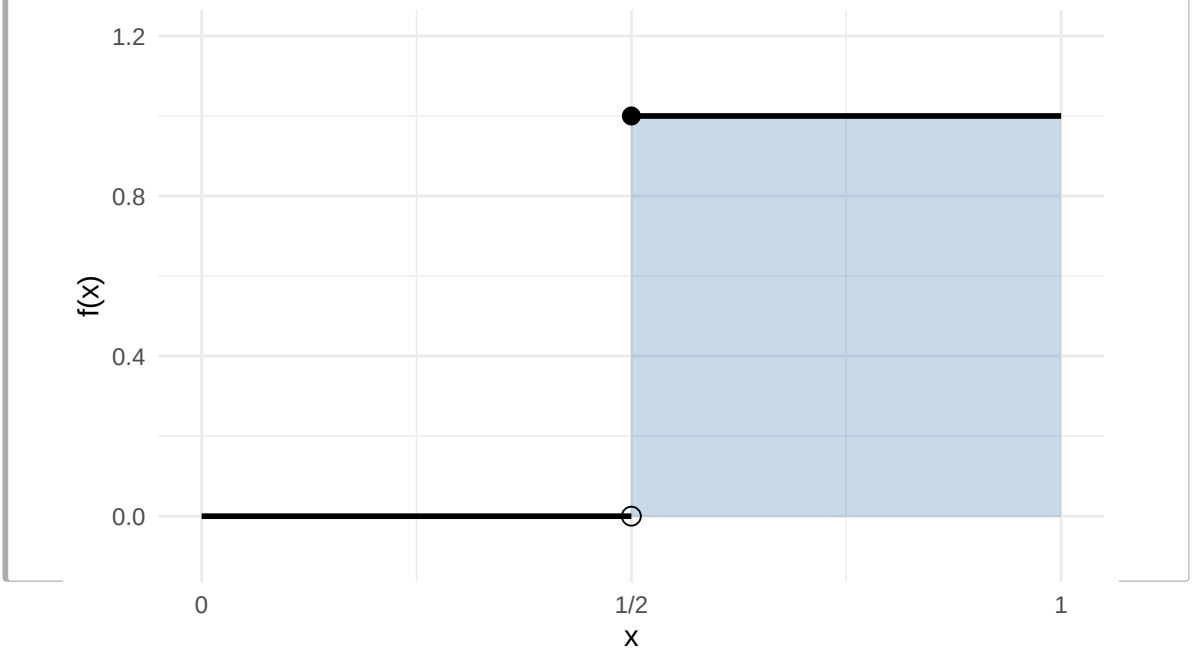


Figure 4: Step function: $f(x) = 0$ on $[0, \frac{1}{2})$ (open circle at the jump) and $f(x) = 1$ on $[\frac{1}{2}, 1]$ (filled circle). The shaded rectangle has area $\frac{1}{2}$, matching the integral computed in Example 0.32.

Together, Theorem 0.13 and Theorem 0.14 establish the chain:

$$\text{differentiable on } [a,b] \;\Rightarrow\; \text{continuous on } [a,b] \;\Rightarrow\; \text{integrable on } [a,b]$$

Example 0.30 and Example 0.32 show that neither implication reverses in general.

Theorem 0.15 (Equal-width Riemann sums converge to the integral)

Theorem 0.15 (Equal-width Riemann sums converge to the integral)**.** *If f is Riemann integrable on $[a,b]$ (Definition 0.23), then for every choice of the sample points x_i^*, the equal-width Riemann sums (Definition 0.25) converge to the integral:*

$$\lim_{n\to\infty} S_n = \int_a^b f(x)\,dx.$$

Proof

Proof. The n equal-width subintervals form a partition of $[a,b]$ whose mesh (Definition 0.20) is $(b-a)/n$, which goes to 0 as $n \to \infty$. So S_n is one of the sums in the limit that defines the integral (Definition 0.22), along a sequence of partitions whose mesh goes to 0, and a limit that has the same value for every choice of partitions has that value along this sequence too. □

Example 0.33 (Equal-width sums for $\int_0^1 x\,dx$)

Example 0.33 (Equal-width sums for $\int_0^1 x\,dx$)**.** Let $f(x) = x$ on $[0,1]$, which is continuous and so Riemann integrable (Theorem 0.14), and take each sample point at the right end of its subinterval, $x_i^* = i/n$. With $\Delta x = 1/n$:

$$\begin{aligned}
S_n &= \sum_{i=1}^{n} \frac{i}{n}\cdot\frac{1}{n} && \text{(equal-width Riemann sum with } x_i^* = i/n\text{)}\\
&= \frac{1}{n^2}\sum_{i=1}^{n} i && \text{(factor out } 1/n^2\text{)}\\
&= \frac{1}{n^2}\cdot\frac{n(n+1)}{2} && \text{(sum of the first } n \text{ integers)}\\
&= \frac{n+1}{2n} && \text{(cancel one factor of } n\text{)}
\end{aligned}$$

So $S_{10} = 0.55$, $S_{100} = 0.505$, $S_{1000} = 0.5005$, and $S_n \to \frac{1}{2}$ as $n \to \infty$. By Theorem 0.15, $\int_0^1 x\,dx = \frac{1}{2}$.

Fundamental Theorem of Calculus

Definition 0.26 (Accumulation function)

Definition 0.26 (Accumulation function)**.** Let f be Riemann integrable on $[a, b]$ (Definition 0.23). The **accumulation function** of f from a is the function F on $[a, b]$ with $F(a) \stackrel{\text{def}}{=} 0$ and

$$F(x) \stackrel{\text{def}}{=} \int_a^x f(t)\,dt \quad \text{for } a < x \leq b.$$

So $F(x)$ is the integral of f accumulated from a up to x. The letter t inside the integral is a placeholder, renamed from x so that x can serve as the upper limit.

Example 0.34 (The accumulation function of a constant)

Example 0.34 (The accumulation function of a constant)**.** Let $f(t) = 2$ on $[0, 3]$, which is Riemann integrable (Example 0.26). For $0 < x \leq 3$, the computation of Example 0.26, with the interval $[0, 3]$ replaced by $[0, x]$, gives

$$\begin{aligned} F(x) &= \int_0^x 2\,dt && \text{(definition of the accumulation function)} \\ &= 2 \cdot (x - 0) && \text{(every Riemann sum is 2 times the total width } x - 0) \\ &= 2x && \text{(subtract)} \end{aligned}$$

and

$$\begin{aligned} F(0) &= 0 \\ &= 2 \cdot 0 \end{aligned}$$

too. For example,

$$\begin{aligned} F(1.5) &= 2 \cdot 1.5 \\ &= 3, \end{aligned}$$

the area of a rectangle of height 2 and width 1.5.

Theorem 0.16 (Fundamental Theorem of Calculus)

Theorem 0.16 (Fundamental Theorem of Calculus)**.** *Let f be a continuous function on a closed interval $[a, b]$.*
Part 1 (Derivative of an integral). *Let $F(x) = \int_a^x f(t)\,dt$ for $x \in [a, b]$ be the accumulation function of f from a (Definition 0.26). Then F is differentiable and:*

$$\frac{\partial}{\partial x}\int_a^x f(t)\,dt = f(x) \tag{3}$$

Note

Continuity on all of $[a,b]$ is a sufficient condition. More generally, Part 1 holds at any individual point x where f is integrable on $[a,b]$ (see Definition 0.23) and continuous at x (see Definition 0.16), even if f has jump discontinuities (Definition 0.17) elsewhere (Rudin 1976, Theorem 6.20, p. 133).

(Larson and Edwards 2018, Theorem 4.11, p. 288)
Part 2 (Evaluation theorem). *The F here may be* any *antiderivative of f — not just the accumulation function from Part 1. If F is an antiderivative of f on $[a,b]$ (i.e., $\frac{\partial}{\partial x}F(x) = f(x)$ for all $x \in [a,b]$), then:*

$$\int_a^b f(x)\,dx = F(b) - F(a) \tag{4}$$

Equivalently, with b replaced by a variable upper limit x, integrating the derivative of F recovers the net change in F:

$$\int_a^x F'(t)\,dt = F(x) - F(a) \tag{5}$$

or equivalently in Leibniz notation:

$$\int_a^x \frac{dF}{dt}\,dt = F(x) - F(a)$$

(Banner 2007, chap. 18; Larson and Edwards 2018, Theorem 4.9, p. 282)

The two parts of the FTC together express that **differentiation and integration are inverse operations**:

- Part 1: differentiating the integral of f recovers f (Equation 3).
- Part 2: the integral of f over $[a,b]$ equals the difference of any antiderivative's values at the endpoints (Equation 4), which rearranges to "integrating the derivative of F recovers the net change in F" (Equation 5).

The standard form of the FTC assumes f is continuous on $[a,b]$; continuity is *sufficient* but not strictly necessary (see the callout note inside Theorem 0.16 for the more general statement). Since differentiability implies continuity (Theorem 0.13), the FTC applies in particular whenever f is differentiable — a common situation in applied statistics.

Definition 0.27 (Evaluation bracket)

Definition 0.27 (Evaluation bracket)**.** For a function F and numbers a and b where F is defined, the **evaluation bracket** is the difference

$$[F(t)]_{t=a}^{t=b} \stackrel{\text{def}}{=} F(b) - F(a).$$

When the variable is clear from the context, it is written $[F(t)]_a^b$. With FTC Part 2 (Theorem 0.16), $\int_a^b f(t)\,dt = [F(t)]_{t=a}^{t=b}$ for any antiderivative F of a continuous f, and computing $F(b) - F(a)$ from the bracket is called **evaluating at the limits** (the limits of integration a and b).

Example 0.35 (Evaluating $\int_1^3 2t\,dt$ with a bracket)

Example 0.35 (Evaluating $\int_1^3 2t\,dt$ with a bracket)**.** $F(t) = t^2$ is an antiderivative of $f(t) = 2t$, since $\frac{\partial}{\partial t} t^2 = 2t$, so

$$\begin{aligned}
\int_1^3 2t\,dt &= [t^2]_{t=1}^{t=3} && \text{(FTC Part 2)} \\
&= 3^2 - 1^2 && \text{(definition of the evaluation bracket)} \\
&= 9 - 1 && \text{(square)} \\
&= 8 && \text{(subtract)}
\end{aligned}$$

Exercise 0.32 (Definite integral of a polynomial)

Exercise 0.32 (Definite integral of a polynomial)**.** Adapted from Miller (2016), Question 1.1.31.
Evaluate the integral:

$$\int_0^1 (x^4 + x^2 + 1)\,dx$$

Solution 0.32

Solution 0.32. Integrate term by term using the power rule for integration:

$$\begin{aligned}\int_0^1 (x^4 + x^2 + 1)\,dx &= \left[\frac{x^5}{5} + \frac{x^3}{3} + x\right]_0^1 \\ &= \left(\frac{1}{5} + \frac{1}{3} + 1\right) - 0 \\ &= \frac{3 + 5 + 15}{15} \\ &= \frac{23}{15}\end{aligned}$$

Exercise 0.33 (Integral of a perfect square)

Exercise 0.33 (Integral of a perfect square)**.** Adapted from Miller (2016), Question 1.1.32.
Evaluate the integral:

$$\int_0^1 (x^2 + 2x + 1)\,dx$$

Solution 0.33

Solution 0.33. Notice that the integrand is a perfect square: $x^2 + 2x + 1 = (x + 1)^2$.
With substitution $u = x + 1$ (where $du = dx$):

$$\begin{aligned}\int_0^1 (x + 1)^2\,dx &= \left[\frac{(x + 1)^3}{3}\right]_0^1 \\ &= \frac{2^3}{3} - \frac{1^3}{3} \\ &= \frac{8 - 1}{3} \\ &= \frac{7}{3}\end{aligned}$$

Alternative (term by term):

$$\begin{aligned}\left[\frac{x^3}{3} + x^2 + x\right]_0^1 &= \frac{1}{3} + 1 + 1 \\ &= \frac{7}{3}\end{aligned}$$

Exercise 0.34 (Integral of a composite power)

Exercise 0.34 (Integral of a composite power)**.** Adapted from Miller (2016), Question 1.1.33.
Evaluate the integral:

$$\int_0^1 (x^2 + 2x + 1)^2 \, dx$$

Solution 0.34

Solution 0.34. Recognize that $x^2 + 2x + 1 = (x + 1)^2$, so the integrand is:

$$\begin{aligned}(x^2 + 2x + 1)^2 &= \left[(x + 1)^2\right]^2 \\ &= (x + 1)^4\end{aligned}$$

Using the substitution $u = x + 1$ with $du = dx$:

$$\begin{aligned}\int_0^1 (x + 1)^4 \, dx &= \left[\frac{(x + 1)^5}{5}\right]_0^1 \\ &= \frac{2^5 - 1^5}{5} \\ &= \frac{32 - 1}{5} \\ &= \frac{31}{5}\end{aligned}$$

Remark: Do not mistakenly write $\int (x^2 + 2x + 1)^2 \, dx = \frac{(x^2+2x+1)^3}{3}$; that formula requires the derivative of the inside function $(2x + 2)$ to be present as a factor in the integrand.

Exercise 0.35 (Integral of odd trigonometric functions)

Exercise 0.35 (Integral of odd trigonometric functions)**.** Adapted from Miller (2016), Question 1.1.34.
Evaluate the integral:

$$\int_{-\pi/2}^{\pi/2} (\sin^3 x \cos x + \sin x \cos x) \, dx$$

Solution 0.35

Solution 0.35. Because $\sin(-x) = -\sin x$ and $\cos(-x) = \cos x$:

$$\sin^3(-x)\cos(-x) + \sin(-x)\cos(-x) = -\sin^3 x \cos x - \sin x \cos x$$

The integrand is an odd function. The integral of any continuous odd function over a symmetric interval $[-a, a]$ is zero:

$$\int_{-\pi/2}^{\pi/2} (\sin^3 x \cos x + \sin x \cos x)\, dx = 0$$

Alternative (by substitution): Set $u = \sin x$, so $du = \cos x\, dx$. As x ranges from $-\pi/2$ to $\pi/2$, u ranges from -1 to 1:

$$\begin{aligned}\int_{-1}^{1} (u^3 + u)\, du &= \left[\frac{u^4}{4} + \frac{u^2}{2}\right]_{-1}^{1} \\ &= \left(\frac{1}{4} + \frac{1}{2}\right) - \left(\frac{1}{4} + \frac{1}{2}\right) \\ &= 0\end{aligned}$$

Exercise 0.36 (Separating even and odd parts of a polynomial)

Exercise 0.36 (Separating even and odd parts of a polynomial)**.** Adapted from Miller (2016), Question 1.1.35.
Evaluate the integral:

$$\int_{-4}^{4} (x^3 + 6x^2 - 2x - 3)\, dx$$

Solution 0.36

Solution 0.36. Split the integrand into its odd part $x^3 - 2x$ and its even part $6x^2 - 3$:

$$\int_{-4}^{4} (x^3 - 2x)\, dx = 0$$

The even part doubles over the half-interval $[0, 4]$:

$$\begin{aligned}\int_{-4}^{4}(6x^2-3)\,dx &= 2\int_{0}^{4}(6x^2-3)\,dx\\ &= 2\big[2x^3-3x\big]_0^4\\ &= 2[2(64)-3(4)]\\ &= 2(128-12)\\ &= 232\end{aligned}$$

Exercise 0.37 (Substitution leading to a natural logarithm)

Exercise 0.37 (Substitution leading to a natural logarithm)**.** Adapted from Miller (2016), Question 1.1.36.
Evaluate the integral:

$$\int_0^1 \frac{x}{1+x^2}\,dx$$

Solution 0.37

Solution 0.37. Let $u = 1+x^2$. Then $du = 2x\,dx$, so $x\,dx = \frac{1}{2}du$.
Transform the limits of integration:

- When $x = 0$:

$$\begin{aligned}u &= 1+0^2\\ &= 1.\end{aligned}$$

- When $x = 1$:

$$\begin{aligned}u &= 1+1^2\\ &= 2.\end{aligned}$$

Applying the substitution:

$$\begin{aligned}\int_0^1 \frac{x}{1+x^2}\,dx &= \frac{1}{2}\int_1^2 \frac{du}{u} \\ &= \frac{1}{2}[\log u]_1^2 \\ &= \frac{1}{2}(\log 2 - \log 1) \\ &= \frac{\log 2}{2}\end{aligned}$$

Remark: When using u-substitution with definite integrals, the transformation $x \mapsto u(x)$ must be one-to-one on the domain of integration.

Exercise 0.38 (Substitution with a polynomial power)

Exercise 0.38 (Substitution with a polynomial power)**.** Adapted from Miller (2016), Question 1.1.37.
Evaluate the integral:

$$\int_0^3 (x^3+3x)^8(x^2+1)\,dx$$

Solution 0.38

Solution 0.38. Let $u = x^3 + 3x$. Then:

$$\begin{aligned} du &= (3x^2+3)\,dx \\ &= 3(x^2+1)\,dx \\ \implies (x^2+1)\,dx &= \frac{du}{3}\end{aligned}$$

Limits:

- When $x = 0$: $u = 0$.
- When $x = 3$:

$$\begin{aligned} u &= 3^3 + 3(3) \\ &= 27 + 9 \\ &= 36.\end{aligned}$$

Substitute:

$$\begin{aligned}\int_0^3 (x^3+3x)^8(x^2+1)\,dx &= \frac{1}{3}\int_0^{36} u^8\,du \\ &= \frac{1}{3}\left[\frac{u^9}{9}\right]_0^{36} \\ &= \frac{36^9}{27}\end{aligned}$$

Exercise 0.39 (Substitution with a trigonometric integrand)

Exercise 0.39 (Substitution with a trigonometric integrand)**.** Adapted from Miller (2016), Question 1.1.38.
Evaluate the integral:

$$\int_0^2 x\cos(3x^2)\,dx$$

Solution 0.39

Solution 0.39. Let $u = 3x^2$. Then $du = 6x\,dx$, meaning $x\,dx = \frac{du}{6}$.
Limits:

- When $x = 0$: $u = 0$.
- When $x = 2$:

$$\begin{aligned}u &= 3(2^2) \\ &= 12.\end{aligned}$$

Substitute:

$$\begin{aligned}\int_0^2 x\cos(3x^2)\,dx &= \frac{1}{6}\int_0^{12} \cos u\,du \\ &= \frac{1}{6}[\sin u]_0^{12} \\ &= \frac{\sin 12}{6}\end{aligned}$$

Exercise 0.40 (Integration by parts with an exponential factor)

Exercise 0.40 (Integration by parts with an exponential factor)**.** Adapted from Miller

(2016), Question 1.1.40.
Evaluate the integral:

$$\int_a^b x^3 \exp\left\{-\frac{x^2}{2}\right\} dx$$

Solution 0.40

Solution 0.40. Factor the integrand as $x^2 \cdot (x \exp\{-x^2/2\})$. Set:

$$u = x^2, \qquad dv = x \exp\left\{-\frac{x^2}{2}\right\} dx$$

Then $du = 2x\,dx$ and $v = -\exp\{-x^2/2\}$.
Using integration by parts ($\int u\,dv = uv - \int v\,du$):

$$\begin{aligned}
\int_a^b x^3 \exp\left\{-\frac{x^2}{2}\right\} dx &= \left[-x^2 \exp\left\{-\frac{x^2}{2}\right\}\right]_a^b - \int_a^b \left[-\exp\left\{-\frac{x^2}{2}\right\}\right](2x\,dx) \\
&= \left[-x^2 \exp\left\{-\frac{x^2}{2}\right\}\right]_a^b + 2\int_a^b x \exp\left\{-\frac{x^2}{2}\right\} dx \\
&= \left[-x^2 \exp\left\{-\frac{x^2}{2}\right\}\right]_a^b + 2\left[-\exp\left\{-\frac{x^2}{2}\right\}\right]_a^b \\
&= \left[-(x^2+2) \exp\left\{-\frac{x^2}{2}\right\}\right]_a^b \\
&= (a^2+2) \exp\left\{-\frac{a^2}{2}\right\} - (b^2+2) \exp\left\{-\frac{b^2}{2}\right\}
\end{aligned}$$

Example 0.36 (FTC Part 1 visualized: accumulation function for $f(t) = 2t$)

Example 0.36 (FTC Part 1 visualized: accumulation function for $f(t) = 2t$)**.** Take $f(t) = 2t$ on $[0, 2]$. The accumulation function from 0 is

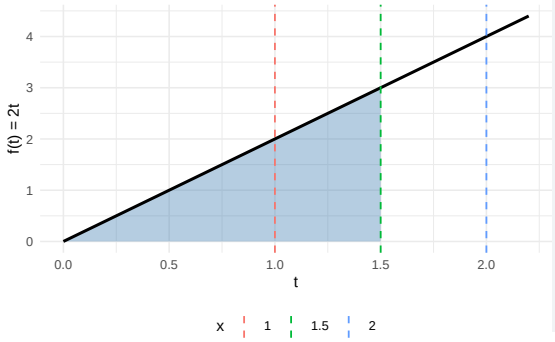
$$\begin{aligned}
F(x) &\overset{\text{def}}{=} \int_0^x 2t\,dt \\
&= \left[t^2\right]_{t=0}^{t=x} \\
&= x^2 - 0^2 \\
&= x^2,
\end{aligned}$$

so $F(x) = x^2$, and indeed

$$\begin{aligned} F'(x) &= 2x \\ &= f(x), \end{aligned}$$

as Theorem 0.16 Part 1 predicts. Figure 5 shows the integrand on the left (shaded area equals $F(x)$ at each x) and the accumulation function $F(x) = x^2$ on the right (its slope at x equals $f(x) = 2x$).

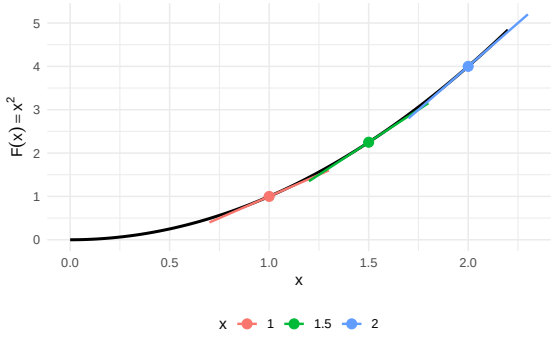
```
ggplot2::ggplot() +
  ggplot2::geom_area(
    data = tibble::tibble(t = seq(0,
    ggplot2::aes(x = t, y = 2 * t),
    fill = "steelblue", alpha = 0.4
  ) +
  ggplot2::geom_function(fun = \(t)
  ggplot2::geom_vline(
    data = tibble::tibble(x = x_mark
    ggplot2::aes(xintercept = x, col
    linetype = "dashed", linewidth =
  ) +
  ggplot2::labs(x = "t", y = "f(t) =
  ggplot2::theme_minimal() +
  ggplot2::theme(legend.position = "
```



(a) $f(t) = 2t$; shaded area equals $F(1.5) = 2.25$; vertical lines mark $x \in \{1, 1.5, 2\}$.

```
slope_df <- tibble::tibble(
  x = x_marks,
  Fx = x_marks^2,
  slope = 2 * x_marks
)

ggplot2::ggplot() +                    ) +
  ggplot2::geom_function(fun = \(x) x^2, xlim = c(0, 2
  ggplot2::geom_point(
    data = slope_df,
    ggplot2::aes(x = x, y = Fx, color = factor(x)),
    size = 3
  ) +
  ggplot2::geom_segment(
    data = slope_df,
    ggplot2::aes(
      x = x - 0.3, xend = x + 0.3,
      y = Fx - 0.3 * slope, yend = Fx + 0.3 * slope,
      color = factor(x)
    ),
    linewidth = 0.8
  ) +
  ggplot2::labs(x = "x", y = expression(F(x) == x^2),
  ggplot2::theme_minimal() +
  ggplot2::theme(legend.position = "bottom")
```



(b) $F(x) = x^2$; tangent slope at each marked x equals $f(x) = 2x$.

Left: $f(t) = 2t$; the shaded area

$$\int_0^{1.5} 2t\,dt = F(1.5) = 2.25;$$

Figure 5: vertical lines mark $x \in \{1, 1.5, 2\}$. Right: $F(x) = x^2$; for each marked x, the tangent slope equals $f(x) = 2x$.

Example 0.37 (CDF and PDF of the exponential distribution)

Example 0.37 (CDF and PDF of the exponential distribution)**.** In what follows, f denotes the PDF and F the CDF — the same letters as the antiderivative pair in Definition 0.13, because the FTC will show F is exactly an antiderivative of f.
Let T be a random variable with the exponential distribution with rate $\lambda > 0$. Its probability density function (PDF) is (Kleinbaum and Klein 2012, sec. II, p. 295, "Survival and Hazard Functions for Selected Distributions"):

$$f(t) = \lambda \mathrm{e}^{-\lambda t}, \quad t \geq 0$$

FTC Part 2 gives the cumulative distribution function (CDF), $F(t) = P(T \leq t)$, from the PDF. Apply the e^{cx} rule from Theorem 0.12 with $c = -\lambda$ to antidifferentiate the integrand:

$$\begin{aligned}
F(t) &= \int_0^t \lambda \mathrm{e}^{-\lambda u}\, du && \text{(the CDF integrates the PDF)} \\
&= \left[\lambda \cdot \frac{1}{-\lambda}\mathrm{e}^{-\lambda u}\right]_{u=0}^{u=t} && \text{(FTC Part 2, with the } \mathrm{e}^{cx} \text{ rule)} \\
&= \left[(-1)\mathrm{e}^{-\lambda u}\right]_{u=0}^{u=t} && (\lambda/(-\lambda) = -1) \\
&= \left[-\mathrm{e}^{-\lambda u}\right]_{u=0}^{u=t} && \text{(multiply by } -1) \\
&= -\mathrm{e}^{-\lambda t} - (-\mathrm{e}^{0}) && \text{(evaluate at the limits)} \\
&= -\mathrm{e}^{-\lambda t} - (-1) && (\mathrm{e}^0 = 1) \\
&= 1 - \mathrm{e}^{-\lambda t} && \text{(rearrange)}
\end{aligned}$$

FTC Part 1 recovers the PDF from the CDF:

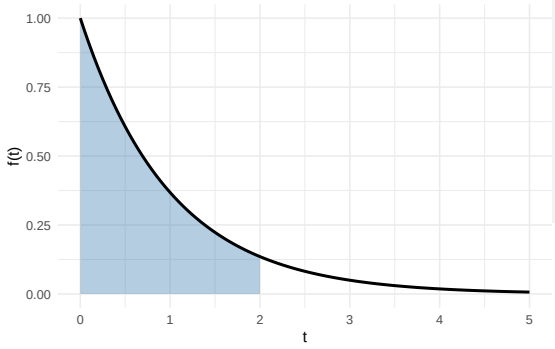
$$\begin{aligned}
\frac{\partial}{\partial t}F(t) &= \frac{\partial}{\partial t}(1 - \mathrm{e}^{-\lambda t}) && \text{(substitute } F) \\
&= \frac{\partial}{\partial t}1 - \frac{\partial}{\partial t}\mathrm{e}^{-\lambda t} && \text{(derivative of a difference)} \\
&= 0 - \frac{\partial}{\partial t}\mathrm{e}^{-\lambda t} && \text{(constant rule)} \\
&= 0 - \mathrm{e}^{-\lambda t} \cdot \frac{\partial}{\partial t}(-\lambda t) && \text{(chain rule, with inner function } -\lambda t) \\
&= 0 - \mathrm{e}^{-\lambda t} \cdot (-\lambda) && \text{(constant multiple rule)} \\
&= \lambda \mathrm{e}^{-\lambda t} && \text{(simplify)} \\
&= f(t) && \text{(definition of } f)
\end{aligned}$$

For a concrete instance: with $\lambda = 1$, the probability that $T \leq 2$ is:

$$\begin{aligned} F(2) &= 1 - \mathrm{e}^{-1 \cdot 2} \\ &= 1 - \mathrm{e}^{-2} \\ &\approx 1 - 0.135 \\ &= 0.865 \end{aligned}$$

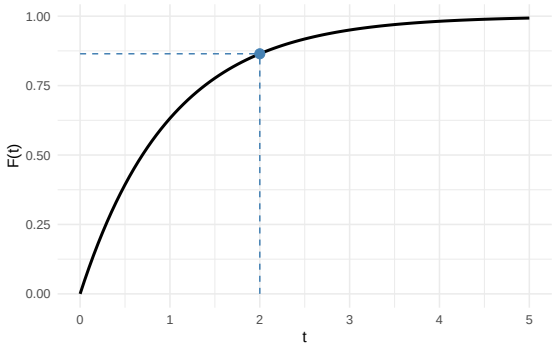
See Figure 6.

```
ggplot2::ggplot() +
  ggplot2::geom_area(
    data = tibble::tibble(t = seq(0,
    ggplot2::aes(x = t, y = lambda *
    fill = "steelblue", alpha = 0.4
  ) +
  ggplot2::geom_function(
    fun = \(t) lambda * exp(-lambda
    xlim = c(0, t_max), linewidth =
  ) +
  ggplot2::labs(x = "t", y = "f(t)")
  ggplot2::theme_minimal()
```



(a) PDF with $\lambda = 1$; shaded area equals $F(2) \approx 0.865$.

```
ggplot2::ggplot() +
  ggplot2::geom_function(
    fun = \(t) 1 - exp(-lambda * t),
    xlim = c(0, t_max), linewidth = 1
  ) +
  ggplot2::geom_point(
    ggplot2::aes(x = t_focus, y = F_at_focus),
    size = 3, color = "steelblue"
  ) +
  ggplot2::geom_segment(
    ggplot2::aes(x = t_focus, xend = t_focus, y = 0, y
    linetype = "dashed", color = "steelblue"
  ) +
  ggplot2::geom_segment(
    ggplot2::aes(x = 0, xend = t_focus, y = F_at_focus
    linetype = "dashed", color = "steelblue"
  ) +
  ggplot2::labs(x = "t", y = "F(t)") +
  ggplot2::theme_minimal()
```



(b) CDF with $\lambda = 1$; point marks $F(2) \approx 0.865$.

Figure 6: Exponential distribution with $\lambda = 1$. Left: the PDF $f(t) = \lambda e^{-\lambda t}$; the shaded area under the curve from 0 to 2 equals $F(2) \approx 0.865$. Right: the CDF $F(t) = 1 - e^{-\lambda t}$; the dashed lines mark the value $F(2)$ computed via FTC Part 2.

Exercise 0.41 (Improper Gaussian-kernel integral)

Exercise 0.41 (Improper Gaussian-kernel integral)**.** Adapted from Miller (2016), Question 1.1.39.

Evaluate the improper integral:

$$\int_0^\infty x \exp\left\{-\frac{x^2}{4}\right\} dx$$

Solution 0.41

Solution 0.41. Let $u = x^2/4$. Then $du = \frac{x}{2}\,dx$, so $x\,dx = 2\,du$.
Limits:

- As $x \to 0$: $u \to 0$.
- As $x \to \infty$: $u \to \infty$.

Substitute:

$$\begin{aligned}\int_0^\infty x \exp\left\{-\frac{x^2}{4}\right\} dx &= 2\int_0^\infty e^{-u}\,du \\ &= 2[-e^{-u}]_0^\infty \\ &= 2(0-(-1)) \\ &= 2\end{aligned}$$

Remark: Integrals of this structure arise directly in calculating moments and normalization constants for Gaussian and Rayleigh probability distributions.

Exercise 0.42 (Convolution of two uniform densities)

Exercise 0.42 (Convolution of two uniform densities)**.** Adapted from Miller (2016), Question 1.1.41.
Let $f(x) = \mathbf{1}_{[0,1]}(x)$ be the indicator of the unit interval:

$$f(x) = \begin{cases} 1 & \text{if } x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

Evaluate the convolution:

$$(f * f)(x) = \int_{-\infty}^{\infty} f(t) f(x-t)\,dt$$

for all $x \in \mathbb{R}$.

Solution 0.42

Solution 0.42. The integrand $f(t)f(x-t)$ equals 1 when both $t \in [0,1]$ and $x-t \in [0,1]$, and equals 0 otherwise.
The condition $0 \leq x-t \leq 1$ rearranges to:

$$x-1 \leq t \leq x$$

For this interval to overlap with $t \in [0,1]$, we must have $x \geq 0$ and $x-1 \leq 1$, meaning $0 \leq x \leq 2$. Outside $[0,2]$, $(f*f)(x) = 0$.
For $x \in [0,2]$, we distinguish two cases:

- **Case 1:** $0 \leq x \leq 1$**.** The overlapping bounds for t are $[0,x]$:

$$\begin{aligned}(f*f)(x) &= \int_0^x 1\,dt \\ &= x\end{aligned}$$

- **Case 2:** $1 \leq x \leq 2$**.** The overlapping bounds for t are $[x-1,1]$:

$$\begin{aligned}(f*f)(x) &= \int_{x-1}^1 1\,dt \\ &= 1-(x-1) \\ &= 2-x\end{aligned}$$

Combining the cases:

$$(f*f)(x) = \begin{cases} x & \text{if } 0 \leq x \leq 1 \\ 2-x & \text{if } 1 < x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Remark: In probability theory, if $X_1, X_2 \sim \text{Uniform}(0,1)$ independently, the probability density function of their sum $S = X_1 + X_2$ is the convolution of their individual densities, producing this symmetric triangular distribution on $[0,2]$.

Videos

Jon Krohn's "Calculus for Machine Learning" YouTube playlist has videos on evaluating definite integrals:

- Definite Integrals
- Definite Integral Exercise

Double Integrals

The **Fubini–Tonelli theorem** states conditions under which the order of integration in a double integral can be exchanged. We state two versions: the Riemann version (Theorem 0.17) is what applied courses usually use for double integrals of continuous functions on simple regions; the σ-finite measure-theoretic version (Theorem 0.18) is included to make the joint-distribution form corollary in *Probability for Data Science* follow from a stated theorem rather than from an aside.

Definition 0.28 (Double integral)

Definition 0.28 (Double integral)**.** Let f be a bounded function on a closed, bounded plane region $R \subseteq \mathbb{R}^2$. Cover R with a grid of rectangles, keep the n rectangles that lie entirely inside R, with areas $\Delta A_1, \dots, \Delta A_n$, and choose a point (x_i, y_i) in the i-th rectangle. The **double integral** of f over R is

$$\iint_R f(x,y)\,dA \overset{\text{def}}{=} \lim_{\|\Delta\|\to 0} \sum_{i=1}^{n} f(x_i, y_i)\,\Delta A_i,$$

where $\|\Delta\|$ is the length of the longest diagonal among the n rectangles, when that limit exists and has the same value for every choice of grids and of the points (x_i, y_i). The symbol dA stands for an element of area.
(Larson and Edwards 2018, sec. 14.2)

Example 0.38 (The double integral of 1 is an area)

Example 0.38 (The double integral of 1 is an area)**.** Let $f(x,y) = 1$ on the rectangle $R = [0,2] \times [0,3]$. Every sum in Definition 0.28 adds up the areas of rectangles inside R, and those sums approach the area of R as the grid gets finer, so

$$\begin{aligned}\iint_R 1\,dA &= 2 \cdot 3 \\ &= 6.\end{aligned}$$

Definition 0.29 (Iterated integral)

Definition 0.29 (Iterated integral)**.** An **iterated integral** is an integral of an integral:

$$\int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx \overset{\text{def}}{=} \int_a^b \left(\int_{g_1(x)}^{g_2(x)} f(x,y)\,dy \right) dx$$

That is, integrate over the inner variable (y) first, holding the outer variable (x) fixed, and then integrate the result over the outer variable. The inner limits $g_1(x)$ and $g_2(x)$ may

depend on the outer variable, or be constants c and d, as in $\int_a^b \int_c^d f(x,y)\,dy\,dx$. Iterated integrals in the other order, $\int\int \cdots\, dx\,dy$, are defined the same way with the roles of x and y swapped.

Example 0.39 (An iterated integral)

Example 0.39 (An iterated integral)**.** Integrating over y first, then x:

$$\begin{aligned}
\int_0^1 \int_0^2 xy\,dy\,dx &= \int_0^1 \left(\int_0^2 xy\,dy \right) dx && \text{(definition of the iterated integral)} \\
&= \int_0^1 x \left(\int_0^2 y\,dy \right) dx && (x \text{ is constant in } y) \\
&= \int_0^1 x \left[\frac{y^2}{2} \right]_{y=0}^{y=2} dx && \text{(antiderivative of } y) \\
&= \int_0^1 2x\,dx && \text{(evaluate at the limits)} \\
&= [x^2]_{x=0}^{x=1} && \text{(antiderivative of } 2x) \\
&= 1 && \text{(evaluate at the limits)}
\end{aligned}$$

Theorem 0.17 (Fubini's theorem (Riemann version))

Theorem 0.17 (Fubini's theorem (Riemann version))**.** *Let f be **continuous** on a plane region $R \subseteq \mathbb{R}^2$.*

1. ***Vertically simple region.*** *If R is defined by $a \le x \le b$ and $g_1(x) \le y \le g_2(x)$, where g_1 and g_2 are continuous on $[a,b]$, then*

$$\iint_R f(x,y)\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\,dy\,dx.$$

2. ***Horizontally simple region.*** *If R is defined by $c \le y \le d$ and $h_1(y) \le x \le h_2(y)$, where h_1 and h_2 are continuous on $[c,d]$, then*

$$\iint_R f(x,y)\,dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x,y)\,dx\,dy.$$

When R can be described both ways, the two iterated integrals are equal — so the order of integration can be exchanged.
(Larson and Edwards 2018, Theorem 14.2, p. 982)

Example 0.40 (Changing the order of integration for a non-rectangular region)

Example 0.40 (Changing the order of integration for a non-rectangular region)**.** Adapted from (Larson and Edwards 2018, sec. 14.2, Example 4, pp. 984–985).
Let X and Y be independent Uniform$(0,1)$ random variables, with joint density $f(x,y) = 1$ on the unit square $[0,1]^2$. Define the function $g(x,y) = \mathrm{e}^{-x^2}\,\mathbb{1}(y \le x)$, where e is Euler's number, and compute its expectation $\mathrm{E}[g(X,Y)]$.
Because the joint density equals 1 on $[0,1]^2$, this expectation is the double integral of g over the unit square:

$$\mathrm{E}[g(X,Y)] = \iint_{[0,1]^2} g(x,y)\,dA.$$

The indicator factor $\mathbb{1}(y \le x)$ equals 1 on the triangular region where $y \le x$ and 0 elsewhere, so only that region, namely $D = \{(x,y) : x \in [0,1],\ y \in [0,x]\}$ (Figure 7), contributes, and there $g(x,y) = \mathrm{e}^{-x^2}$:

$$\mathrm{E}[g(X,Y)] = \iint_{D} \mathrm{e}^{-x^2}\,dA.$$

```
region <- tibble::tibble(x = c(0, 1, 1), y = c(0, 0, 1))
ggplot2::ggplot(region, ggplot2::aes(x = x, y = y)) +
  ggplot2::geom_polygon(
    fill = "steelblue", alpha = 0.4, color = "black", linewidth = 0.7
  ) +
  ggplot2::annotate(
    "text", x = 0.65, y = 0.28, label = "D", size = 8, fontface = "italic"
  ) +
  ggplot2::annotate(
    "text", x = 0.36, y = 0.52, label = "y == x",
    parse = TRUE, angle = 45
  ) +
  ggplot2::annotate(
    "text", x = 1.08, y = 0.5, label = "x == 1",
    parse = TRUE, angle = 90, hjust = 0.5
  ) +
  ggplot2::annotate(
    "text", x = 0.5, y = -0.1, label = "y == 0", parse = TRUE
  ) +
  ggplot2::scale_x_continuous(breaks = c(0, 0.5, 1), limits = c(-0.1, 1.3)) +
  ggplot2::scale_y_continuous(breaks = c(0, 0.5, 1), limits = c(-0.2, 1.2)) +
  ggplot2::coord_equal() +
  ggplot2::labs(x = "x", y = "y") +
  ggplot2::theme_minimal()
```

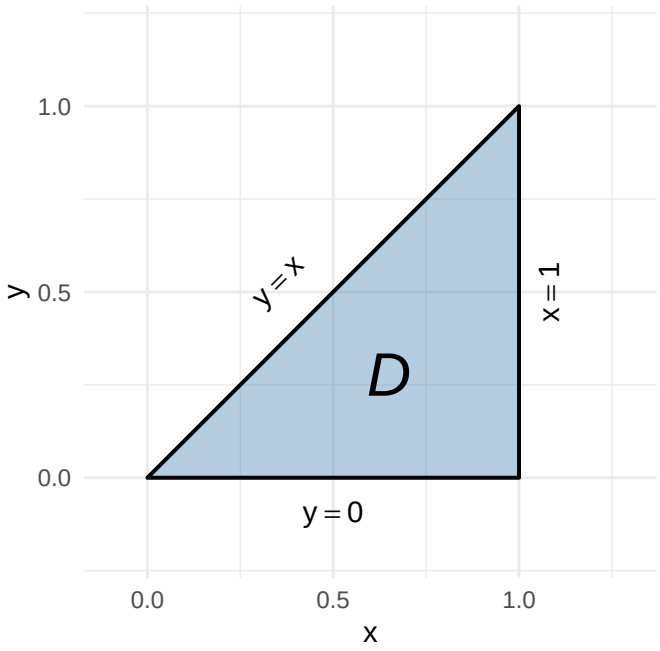


Figure 7: Triangular integration region $D = \{(x, y) : x \in [0, 1],\ y \in [0, x]\}$, bounded below by $y = 0$, above-left by $y = x$, and on the right by $x = 1$.

Order $dx\,dy$ is intractable. Re-describing D as $D = \{(x, y) : y \in [0, 1],\ x \in [y, 1]\}$, the inner integral is

$$\int_y^1 \mathrm{e}^{-x^2}\,dx,$$

which cannot be computed with FTC Part 2: no antiderivative of e^{-x^2} can be written as a finite formula built from powers, exponentials, logarithms, and trigonometric functions (an **elementary** antiderivative).

Order $dy\,dx$ works. Applying Theorem 0.17 Part 1 (e^{-x^2} is continuous and D is the vertically simple region $x \in [0, 1]$, $y \in [0, x]$):

$$\begin{aligned}
\iint_D \mathrm{e}^{-x^2}\,dA &= \int_0^1 \int_0^x \mathrm{e}^{-x^2}\,dy\,dx && \text{(Fubini, vertically simple region)} \\
&= \int_0^1 \mathrm{e}^{-x^2}\left(\int_0^x dy\right) dx && (\mathrm{e}^{-x^2} \text{ is constant in } y) \\
&= \int_0^1 x\,\mathrm{e}^{-x^2}\,dx && \left(\textstyle\int_0^x dy = x\right)
\end{aligned}$$

To antidifferentiate $x\,\mathrm{e}^{-x^2}$, let $u = x^2$ be the inner function. By the chain rule (Theorem 0.10), with $\frac{du}{dx} = 2x$,

$$\begin{aligned}
\frac{d}{dx}\left(-\frac{1}{2}\,\mathrm{e}^{-u}\right) &= -\frac{1}{2}\,\mathrm{e}^{-u}\cdot(-1)\cdot\frac{du}{dx} && \text{(chain rule)} \\
&= -\frac{1}{2}\,\mathrm{e}^{-x^2}\cdot(-1)\cdot 2x && (\text{substitute } u = x^2 \text{ and } du/dx = 2x) \\
&= x\,\mathrm{e}^{-x^2} && \text{(multiply)}
\end{aligned}$$

so $-\frac{1}{2}\,\mathrm{e}^{-x^2}$ is an antiderivative of $x\,\mathrm{e}^{-x^2}$, and

$$\begin{aligned}
\int_0^1 x\,\mathrm{e}^{-x^2}\,dx &= \left[-\frac{1}{2}\,\mathrm{e}^{-x^2}\right]_0^1 && \text{(FTC Part 2)} \\
&= -\frac{1}{2}(\mathrm{e}^{-1} - \mathrm{e}^{0}) && \text{(evaluate at the limits)} \\
&= -\frac{1}{2}(\mathrm{e}^{-1} - 1) && (\mathrm{e}^0 = 1) \\
&= \frac{1 - \mathrm{e}^{-1}}{2} && (\text{distribute } -\tfrac{1}{2}) \\
&= \frac{e - 1}{2e} && (\text{multiply numerator and denominator by } e) \\
&\approx 0.316
\end{aligned}$$

The solid whose volume equals this integral is shown in Figure 8.

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 8: Surface $z = e^{-x^2}$ over the region $D = \{(x, y) : x \in [0, 1],\ y \in [0, x]\}$. The surface depends only on x (constant in y), so for each x the inner integral over $y \in [0, x]$ contributes $x \cdot e^{-x^2}$.

Exercise 0.43 (Double integral over a triangular region)

Exercise 0.43 (Double integral over a triangular region)**.** Adapted from Miller (2016), Question 1.1.50.
Evaluate the integral:

$$\int_0^1 \int_0^x xy \, dy \, dx$$

and confirm the result by reversing the order of integration.

Solution 0.43

Solution 0.43. **1. Given order (y then x):**

$$\begin{aligned} \int_0^x xy \, dy &= x \left[\frac{y^2}{2} \right]_0^x \\ &= \frac{x^3}{2} \end{aligned}$$

$$\begin{aligned} \int_0^1 \frac{x^3}{2} \, dx &= \left[\frac{x^4}{8} \right]_0^1 \\ &= \frac{1}{8} \end{aligned}$$

2. Reversed order (x then y): The triangular region $T = \{(x, y) : 0 \le x \le 1, 0 \le y \le x\}$ is equivalently described by $T = \{(x, y) : 0 \le y \le 1, y \le x \le 1\}$:

$$\begin{aligned} \int_0^1 \int_y^1 xy \, dx \, dy &= \int_0^1 y \left[\frac{x^2}{2} \right]_{x=y}^1 dy \\ &= \int_0^1 \frac{y(1 - y^2)}{2} \, dy \end{aligned}$$

$$\int_0^1 \left(\frac{y}{2} - \frac{y^3}{2}\right) dy = \left[\frac{y^2}{4} - \frac{y^4}{8}\right]_0^1$$
$$= \frac{1}{4} - \frac{1}{8}$$
$$= \frac{1}{8}$$

Both orders yield 1/8.

Exercise 0.44 (Switching order of integration on a non-rectangular region)

Exercise 0.44 (Switching order of integration on a non-rectangular region)**.** Adapted from Miller (2016), Question 1.1.51.
Express the integral:

$$\int_0^1 \int_0^x ye^{-xy}\, dy\, dx$$

by reversing the order of integration.

Solution 0.44

Solution 0.44. In the given order, the inner integral $\int_0^x ye^{-xy}\, dy$ requires integration by parts with respect to y.
Switching the order over the triangular domain $\{(x, y) : 0 \leq y \leq 1, y \leq x \leq 1\}$:

$$\int_0^1 \int_y^1 ye^{-xy}\, dx\, dy$$

Now the inner integral has the factor y in place for direct integration with respect to x:

$$\int_y^1 ye^{-xy}\, dx = [-e^{-xy}]_{x=y}^1$$
$$= e^{-y^2} - e^{-y}$$

The double integral becomes:

$$\int_0^1 (e^{-y^2} - e^{-y})\, dy = \int_0^1 e^{-y^2}\, dy - [-e^{-y}]_0^1$$
$$= \int_0^1 e^{-y^2}\, dy - (1 - e^{-1})$$

The term $\int_0^1 e^{-y^2}\,dy$ has no elementary antiderivative; it can be written using the error function $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}}\int_0^z e^{-t^2}\,dt$ as $\frac{\sqrt{\pi}}{2}\operatorname{erf}(1) \approx 0.7468$. The overall value is:

$$\frac{\sqrt{\pi}}{2}\operatorname{erf}(1) + e^{-1} - 1 \approx 0.1147$$

Example 0.41 (When conditions fail: a counterexample)

Example 0.41 (When conditions fail: a counterexample)**.** The conditions in Theorem 0.17 are not merely technical — when they fail, iterated integrals can exist yet disagree.
Let

$$f(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2}$$

on the unit square $R = [0,1]\times[0,1]$. Strictly, f is defined on $R\setminus\{(0,0)\}$: the denominator vanishes at the origin, so f is undefined there (we return to this point in the condition check).
Integrating y first, then x:
Holding x fixed and differentiating in y (by the quotient rule, Theorem 0.9),

$$\begin{aligned}\frac{d}{dy}\frac{y}{x^2+y^2} &= \frac{(x^2+y^2) - y\cdot 2y}{(x^2+y^2)^2}\\ &= \frac{x^2-y^2}{(x^2+y^2)^2}.\end{aligned}$$

(A derivative in one variable with the others held fixed is a partial derivative, defined on the vector calculus page.) The arctangent arctan is the inverse of the tangent function from trigonometry; all this example needs is that $\frac{d}{dx}\arctan(x) = \frac{1}{1+x^2}$, $\arctan(0) = 0$, and $\arctan(1) = \frac{\pi}{4}$.

$$\begin{aligned}\int_0^1\int_0^1 f(x,y)\,dy\,dx &= \int_0^1 \left[\frac{y}{x^2+y^2}\right]_{y=0}^{y=1} dx\\ &= \int_0^1 \frac{1}{x^2+1}\,dx\\ &= [\arctan(x)]_0^1\\ &= \frac{\pi}{4}\end{aligned}$$

Integrating x first, then y:
Holding y fixed and differentiating in x,

$$\begin{aligned}\frac{d}{dx}\left(-\frac{x}{x^2+y^2}\right) &= -\frac{(x^2+y^2)-x\cdot 2x}{(x^2+y^2)^2}\\ &= \frac{x^2-y^2}{(x^2+y^2)^2}\,:\end{aligned}$$

$$\begin{aligned}\int_0^1\int_0^1 f(x,y)\,dx\,dy &= \int_0^1\left[-\frac{x}{x^2+y^2}\right]_{x=0}^{x=1}dy\\ &= \int_0^1\left(-\frac{1}{1+y^2}\right)dy\\ &= -[\arctan(y)]_0^1\\ &= -\frac{\pi}{4}\end{aligned}$$

Conclusion: $\frac{\pi}{4}\neq-\frac{\pi}{4}$, so the two iterated integrals are unequal. Theorem 0.17 does not apply here.

Why Theorem 0.17's condition fails: Theorem 0.17 requires f to be **continuous** on R. The denominator $(x^2+y^2)^2$ vanishes at the origin $(0,0)\in R$, so f is *not even defined* there — let alone continuous — and the theorem does not apply.

(Wikipedia contributors 2024)

Figure 9 shows the surface. The failure comes from the origin, where f is undefined and unbounded.

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 9: Surface $f(x,y)=(x^2-y^2)/(x^2+y^2)^2$ on $[0,1]^2$, sampled away from the origin and clipped to $[-50,50]$ for display. Approaching the origin, the function grows without bound along the x-axis (red ridge, $f>0$ when $|x|>|y|$) and falls without bound along the y-axis (blue ridge, $f<0$ when $|y|>|x|$). Because f is undefined at $(0,0)$, f is not continuous on R and Theorem 0.17 does not apply.

Corollary 0.2 (Continuous functions on a rectangle (corollary of Theorem 0.17))

Corollary 0.2 (Continuous functions on a rectangle (corollary of Theorem 0.17))**.** *If $f:[a,b]\times[c,d]\to\mathbb{R}$ is **continuous** on the closed bounded rectangle $[a,b]\times[c,d]$, then:*

$$\begin{aligned}\int_a^b\left(\int_c^d f(x,y)\,dy\right)dx &= \int_c^d\left(\int_a^b f(x,y)\,dx\right)dy\\ &= \iint_{[a,b]\times[c,d]} f(x,y)\,dA.\end{aligned}$$

(Larson and Edwards 2018, Theorem 14.2, p. 982)

Proof

Proof. A closed bounded rectangle $[a,b]\times[c,d]$ is both vertically simple (with $g_1 \equiv c$, $g_2 \equiv d$) and horizontally simple (with $h_1 \equiv a$, $h_2 \equiv b$). Applying both parts of Theorem 0.17 to f on this rectangle gives the two iterated forms shown. □

Example 0.42 (Evaluating a double integral on a rectangle)

Example 0.42 (Evaluating a double integral on a rectangle)**.** Structure adapted from (Larson and Edwards 2018, sec. 14.2, Example 2, pp. 982–983); the integrand x^2+y^2 is original, chosen so the integral equals $\mathrm{E}[g(X,Y)]$ for $g(x,y)=x^2+y^2$.
Let X and Y be independent Uniform$(0,1)$ random variables, with joint density $f(x,y)=1$ on the unit square $R=\{(x,y): x\in[0,1],\ y\in[0,1]\}$ (Figure 10). Define the function $g(x,y)=x^2+y^2$, and compute its expectation $\mathrm{E}[g(X,Y)]$.
Because the joint density equals 1 on R, this expectation is the double integral of g over R:

$$\mathrm{E}[g(X,Y)] = \iint_R (x^2+y^2)\,dA.$$

```r
ggplot2::ggplot() +
  ggplot2::annotate(
    "rect", xmin = 0, xmax = 1, ymin = 0, ymax = 1,
    fill = "steelblue", alpha = 0.4, color = "black", linewidth = 0.7
  ) +
  ggplot2::annotate(
    "text", x = 0.5, y = 0.5, label = "R", size = 8, fontface = "italic"
  ) +
  ggplot2::scale_x_continuous(breaks = c(0, 1), limits = c(-0.15, 1.25)) +
  ggplot2::scale_y_continuous(breaks = c(0, 1), limits = c(-0.15, 1.25)) +
  ggplot2::coord_equal() +
  ggplot2::labs(x = "x", y = "y") +
  ggplot2::theme_minimal()
```

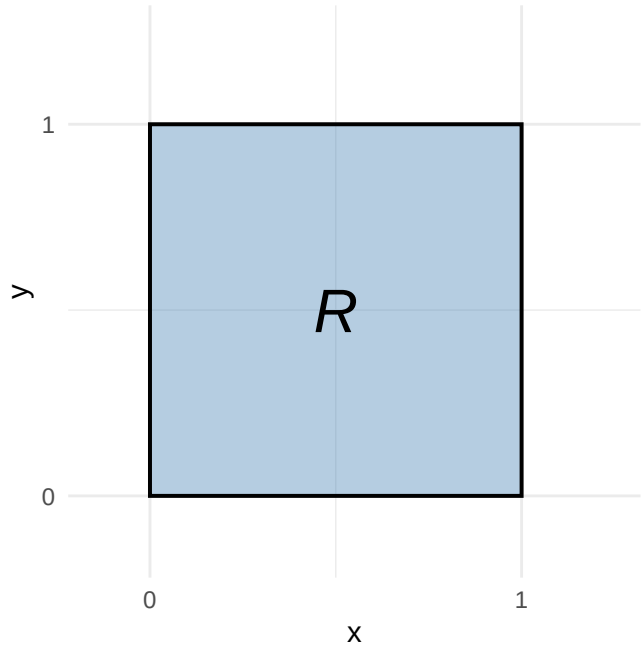


Figure 10: Integration region $R = [0,1]^2$, the unit square.

The integrand is continuous on R, so Corollary 0.2 applies and either order of integration yields the same value.

Integrating y first, then x:

$$\begin{aligned}
\mathrm{E}[g(X,Y)] &= \int_0^1 \int_0^1 (x^2+y^2)\,dy\,dx \\
&= \int_0^1 \left[x^2 y + \frac{y^3}{3}\right]_0^1 dx \\
&= \int_0^1 \left(x^2 + \frac{1}{3}\right) dx \\
&= \left[\frac{x^3}{3} + \frac{x}{3}\right]_0^1 \\
&= \frac{2}{3}
\end{aligned}$$

Integrating x first, then y (verifying the order can be swapped):

$$\begin{aligned}
\int_0^1 \int_0^1 (x^2+y^2)\,dx\,dy &= \int_0^1 \left[\frac{x^3}{3} + y^2 x\right]_0^1 dy \\
&= \int_0^1 \left(\frac{1}{3} + y^2\right) dy \\
&= \left[\frac{y}{3} + \frac{y^3}{3}\right]_0^1 \\
&= \frac{2}{3}
\end{aligned}$$

Both orders give $\frac{2}{3}$, as Corollary 0.2 guarantees.
As a cross-check, linearity of expectation gives the same value: since

$$\begin{aligned}
\mathrm{E}[X^2] &= \int_0^1 x^2\,dx \\
&= \frac{1}{3}
\end{aligned}$$

for $X \sim \mathrm{Uniform}(0,1)$ (and likewise for Y),

$$\begin{aligned}
\mathrm{E}[g(X,Y)] &= \mathrm{E}[X^2 + Y^2] \\
&= \mathrm{E}[X^2] + \mathrm{E}[Y^2] \\
&= \frac{1}{3} + \frac{1}{3} \\
&= \frac{2}{3}.
\end{aligned}$$

The solid whose volume equals this integral is shown in Figure 11.

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 11: Surface $z = x^2 + y^2$ over the unit square $[0,1]^2$. The double integral $\frac{2}{3}$ is the volume between this surface and the xy-plane, and equals $\mathrm{E}[X^2 + Y^2]$.

Exercise 0.45 (Double integral of a polynomial over a rectangle)

Exercise 0.45 (Double integral of a polynomial over a rectangle)**.** Adapted from Miller (2016), Question 1.1.47.
Evaluate the double integral:

$$\int_0^2 \int_0^3 5(x^2y + xy^2 + 2)\, dy\, dx$$

Solution 0.45

Solution 0.45. Integrate with respect to y first:

$$\begin{aligned}\int_0^3 (x^2y + xy^2 + 2)\, dy &= \left[\frac{x^2y^2}{2} + \frac{xy^3}{3} + 2y\right]_{y=0}^{3} \\ &= \frac{9}{2}x^2 + 9x + 6\end{aligned}$$

Now integrate with respect to x:

$$\begin{aligned}5\int_0^2 \left(\frac{9}{2}x^2 + 9x + 6\right) dx &= 5\left[\frac{3}{2}x^3 + \frac{9}{2}x^2 + 6x\right]_0^2 \\ &= 5\left[\frac{3}{2}(8) + \frac{9}{2}(4) + 6(2)\right] \\ &= 5[12 + 18 + 12] \\ &= 5(42) \\ &= 210\end{aligned}$$

(Note: The source text carried an arithmetic slip in the inner coefficient resulting in 190*; the exact value is* 210*.)*

Exercise 0.46 (Choosing the order of integration)

Exercise 0.46 (Choosing the order of integration)**.** Adapted from Miller (2016), Question 1.1.48.
Evaluate the double integral:

$$\int_0^6 \int_0^5 xe^{-xy}\,dy\,dx$$

Solution 0.46

Solution 0.46. Integrating with respect to x first requires integration by parts. Integrating with respect to y first is much simpler because the factor of x is already present:

$$\begin{aligned}\int_0^5 xe^{-xy}\,dy &= [-e^{-xy}]_{y=0}^5 \\ &= 1 - e^{-5x}\end{aligned}$$

Now integrate with respect to x:

$$\begin{aligned}\int_0^6 (1 - e^{-5x})\,dx &= \left[x + \frac{1}{5}e^{-5x}\right]_0^6 \\ &= \left(6 + \frac{1}{5}e^{-30}\right) - \left(0 + \frac{1}{5}\right) \\ &= \frac{29}{5} + \frac{e^{-30}}{5}\end{aligned}$$

Exercise 0.47 (Product of single integrals for separable functions)

Exercise 0.47 (Product of single integrals for separable functions)**.** Adapted from Miller (2016), Question 1.1.49.
Let $m, n > 0$. Evaluate:

$$\int_0^1 \int_0^1 x^m y^n\,dy\,dx$$

Solution 0.47

Solution 0.47. Because the region of integration is a rectangle $[0,1] \times [0,1]$ and the integrand factors as $g(x)h(y) = x^m \cdot y^n$, the double integral splits into the product of two single-variable integrals:

$$\int_0^1 \int_0^1 x^m y^n\,dy\,dx = \left(\int_0^1 x^m\,dx\right)\left(\int_0^1 y^n\,dy\right)$$

Evaluating each factor:

$$\int_0^1 x^m\,dx = \frac{1}{m+1}, \qquad \int_0^1 y^n\,dy = \frac{1}{n+1}$$

Therefore:

$$\int_0^1 \int_0^1 x^m y^n\,dy\,dx = \frac{1}{(m+1)(n+1)}$$

Remark: When joint random variables X and Y are independent, their joint density factors as $f_{X,Y}(x,y) = f_X(x)f_Y(y)$, and joint probabilities over product sets factor in exactly this way.

Exercise 0.48 (Double integral with polynomial and radical terms)

Exercise 0.48 (Double integral with polynomial and radical terms)**.** Adapted from Miller (2016), Question 1.1.52.
Evaluate the double integral:

$$\int_0^1 \int_0^1 (x^2 + 2xy + y\sqrt{x})\,dy\,dx$$

Solution 0.48

Solution 0.48. Integrate with respect to y first:

$$\begin{aligned}\int_0^1 (x^2 + 2xy + y\sqrt{x})\,dy &= \left[x^2 y + xy^2 + \frac{y^2\sqrt{x}}{2}\right]_{y=0}^{1} \\ &= x^2 + x + \frac{1}{2}x^{1/2}\end{aligned}$$

Now integrate with respect to x:

$$\begin{aligned}\int_0^1 \left(x^2 + x + \frac{1}{2}x^{1/2}\right)dx &= \left[\frac{x^3}{3} + \frac{x^2}{2} + \frac{1}{2}\cdot\frac{x^{3/2}}{3/2}\right]_0^1 \\ &= \left[\frac{x^3}{3} + \frac{x^2}{2} + \frac{1}{3}x^{3/2}\right]_0^1 \\ &= \frac{1}{3} + \frac{1}{2} + \frac{1}{3} \\ &= \frac{2}{3} + \frac{1}{2} \\ &= \frac{7}{6}\end{aligned}$$

Exercise 0.49 (Double integral of an affine function)

Exercise 0.49 (Double integral of an affine function)**.** Adapted from Miller (2016), Question 1.1.53.
Let a, b, c be constants. Evaluate:

$$\int_0^1 \int_0^1 (ax + by + c)\, dy\, dx$$

Solution 0.49

Solution 0.49. By linearity of the integral:

$$\begin{aligned}\int_0^1 \int_0^1 (ax + by + c)\, dy\, dx &= a \int_0^1 x\, dx \int_0^1 1\, dy + b \int_0^1 1\, dx \int_0^1 y\, dy + c \int_0^1 1\, dx \int_0^1 1\, dy \\ &= a\left(\frac{1}{2}\right)(1) + b(1)\left(\frac{1}{2}\right) + c(1)(1) \\ &= \frac{a}{2} + \frac{b}{2} + c\end{aligned}$$

Exercise 0.50 (Harmonic numbers via double integrals)

Exercise 0.50 (Harmonic numbers via double integrals)**.** Adapted from Miller (2016), Question 1.1.54.
Prove that for every positive integer n:

$$\begin{aligned}\int_0^1 \int_0^1 n(1 - xy)^{n-1}\, dx\, dy &= \sum_{k=1}^{n} \frac{1}{k} \\ &= 1 + \frac{1}{2} + \cdots + \frac{1}{n}\end{aligned}$$

Solution 0.50

Solution 0.50. **Method 1 (integration then geometric series):** Integrate with respect to x first:

$$\int_0^1 n(1-xy)^{n-1}\,dx = \left[-\frac{(1-xy)^n}{y}\right]_{x=0}^{1}$$
$$= \frac{1-(1-y)^n}{y}$$

Use the finite geometric series identity $\sum_{k=0}^{n-1} r^k = \frac{1-r^n}{1-r}$ with $r = 1-y$:

$$\frac{1-(1-y)^n}{y} = \sum_{k=0}^{n-1}(1-y)^k$$

Now integrate each term with respect to y from 0 to 1:

$$\int_0^1 (1-y)^k\,dy = \left[-\frac{(1-y)^{k+1}}{k+1}\right]_0^1$$
$$= \frac{1}{k+1}$$

Summing over $k = 0, 1, \ldots, n-1$:

$$\sum_{k=0}^{n-1} \frac{1}{k+1} = \sum_{j=1}^{n} \frac{1}{j}$$
$$= 1 + \frac{1}{2} + \cdots + \frac{1}{n}$$

Method 2 (binomial expansion): Expand $(1-xy)^{n-1} = \sum_{k=0}^{n-1} \binom{n-1}{k}(-1)^k (xy)^k$. Integrating over $[0,1]^2$:

$$\int_0^1 \int_0^1 (xy)^k\,dx\,dy = \left(\frac{1}{k+1}\right)^2$$

Multiplying by n and using the identity $\frac{n}{k+1}\binom{n-1}{k} = \binom{n}{k+1}$:

$$\int_0^1 \int_0^1 n(1-xy)^{n-1}\,dx\,dy = \sum_{k=0}^{n-1} \binom{n}{k+1} \frac{(-1)^k}{k+1}$$
$$= \sum_{j=1}^{n} \frac{1}{j}$$

Remark: This integral identity connects multivariable integration with harmonic numbers $H_n = \sum_{k=1}^{n} \frac{1}{k} \approx \log n + \gamma$, where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant.

Theorem 0.18 (Fubini–Tonelli theorem (measure-theoretic form))

Theorem 0.18 (Fubini–Tonelli theorem (measure-theoretic form))**.** *Let* $(\Omega_1, \mathcal{F}_1, \mu_1)$

and $(\Omega_2, \mathcal{F}_2, \mu_2)$ be measure spaces with σ-finite measures, and let $f : \Omega_1 \times \Omega_2 \to \mathbb{R}$ be measurable with respect to the product σ-algebra $\mathcal{F}_1 \otimes \mathcal{F}_2$. If either

*(a) $f \geq 0$ almost everywhere with respect to the product measure $\mu_1 \otimes \mu_2$ (**Tonelli's theorem**), or*

*(b) $\int_{\Omega_1 \times \Omega_2} |f| \, d(\mu_1 \otimes \mu_2) < \infty$, that is, f is absolutely integrable (**Fubini's theorem**),*

then both iterated integrals exist, agree with the double integral, and equal each other:

$$\begin{aligned}\int_{\Omega_1 \times \Omega_2} f \, d(\mu_1 \otimes \mu_2) &= \int_{\Omega_1} \left(\int_{\Omega_2} f(\omega_1, \omega_2) \, d\mu_2(\omega_2) \right) d\mu_1(\omega_1) \\ &= \int_{\Omega_2} \left(\int_{\Omega_1} f(\omega_1, \omega_2) \, d\mu_1(\omega_1) \right) d\mu_2(\omega_2).\end{aligned}$$

(Billingsley 1995, Theorem 18.3; Gut 2013, Theorem 9.1, p. 65; Fubini 1907; Wikipedia contributors 2024)

Remark 0.3 (Fubini–Tonelli for probability measures)

Remark 0.3 (Fubini–Tonelli for probability measures). Applied courses rarely need the measure-theoretic generalization itself, but it is what justifies the joint-distribution form corollary in *Probability for Data Science*. A probability measure P on Ω has $P(\Omega) = 1 < \infty$, so it is finite, and hence σ-finite (finite and σ-finite measures); for probability measures, the σ-finiteness condition is automatic.

The integrability conditions (nonnegativity or absolute integrability) still need to be verified in each application. For example, Lebesgue measure (ordinary length) on $[0, 1]$ is a probability measure, so the σ-finiteness condition holds for both factors $[0, 1]$, yet the two iterated integrals in Example 0.41 are $\pi/4$ and $-\pi/4$. So σ-finiteness alone does not make the iterated integrals agree.

Example 0.43 (Positive application of Theorem 0.18)

Example 0.43 (Positive application of Theorem 0.18)**.** Let X and Y be independent Exponential(1) random variables, with joint density $f(x, y) = e^{-(x+y)}$ for $x, y \geq 0$.

The probability $P(X \leq 1, Y \leq 1)$ is the integral of f over $[0, 1]^2$ with respect to Lebesgue measure (ordinary length) in each coordinate. Lebesgue measure on $[0, \infty)$ is σ-finite, because $[0, \infty)$ is the union of the intervals $[0, n]$, $n \in \mathbb{N}$, each of finite length n; so the σ-finiteness condition of Theorem 0.18 holds. Since $f(x, y) = e^{-(x+y)} \geq 0$, condition (a) (Tonelli's theorem, nonnegativity) is also satisfied.

By Theorem 0.18, both iterated integrals exist and agree. Integrating y first, then x:

$$
\begin{aligned}
P(X \le 1,\, Y \le 1) &= \int_0^1 \int_0^1 e^{-(x+y)}\, dy\, dx \\
&= \int_0^1 \int_0^1 e^{-x} e^{-y}\, dy\, dx && \text{(exponential of a sum)} \\
&= \int_0^1 e^{-x} \left(\int_0^1 e^{-y}\, dy \right) dx && (e^{-x} \text{ is constant in } y) \\
&= \int_0^1 e^{-x} [-e^{-y}]_{y=0}^{y=1}\, dx && \text{(antiderivative of } e^{-y}) \\
&= \int_0^1 e^{-x}(1 - e^{-1})\, dx && \text{(evaluate at the limits)} \\
&= (1 - e^{-1}) \int_0^1 e^{-x}\, dx && (1 - e^{-1} \text{ is constant in } x) \\
&= (1 - e^{-1})[-e^{-x}]_{x=0}^{x=1} && \text{(antiderivative of } e^{-x}) \\
&= (1 - e^{-1})^2 && \text{(evaluate at the limits)}
\end{aligned}
$$

Integrating x first, then y:

$$
\begin{aligned}
P(X \le 1,\, Y \le 1) &= \int_0^1 \int_0^1 e^{-(x+y)}\, dx\, dy \\
&= \int_0^1 \int_0^1 e^{-y} e^{-x}\, dx\, dy && \text{(exponential of a sum)} \\
&= \int_0^1 e^{-y} \left(\int_0^1 e^{-x}\, dx \right) dy && (e^{-y} \text{ is constant in } x) \\
&= \int_0^1 e^{-y} [-e^{-x}]_{x=0}^{x=1}\, dy && \text{(antiderivative of } e^{-x}) \\
&= \int_0^1 e^{-y}(1 - e^{-1})\, dy && \text{(evaluate at the limits)} \\
&= (1 - e^{-1}) \int_0^1 e^{-y}\, dy && (1 - e^{-1} \text{ is constant in } y) \\
&= (1 - e^{-1})[-e^{-y}]_{y=0}^{y=1} && \text{(antiderivative of } e^{-y}) \\
&= (1 - e^{-1})^2 && \text{(evaluate at the limits)}
\end{aligned}
$$

Both iterated integrals equal $(1 - e^{-1})^2 \approx 0.400$, as Theorem 0.18 guarantees when condition (a) holds.

Example 0.44 (When neither Fubini–Tonelli condition is satisfied)

Example 0.44 (When neither Fubini–Tonelli condition is satisfied)**.** The same function $f(x,y) = (x^2-y^2)/(x^2+y^2)^2$ from Example 0.41 illustrates a case where neither condition of Theorem 0.18 is satisfied.

Why Theorem 0.18's conditions fail: $\iint_R |f|\,dA = \infty$, which violates condition (b). Switching to polar coordinates (r,θ) near the origin, the integrand satisfies

$$\begin{aligned} |f(x,y)| &= \left|x^2-y^2\right|/(x^2+y^2)^2 \\ &= |\cos 2\theta|/r^2, \end{aligned}$$

so

$$\begin{aligned} \iint_R |f|\,dA &\geq \int_0^{\pi/2}\int_0^{\varepsilon} \frac{|\cos 2\theta|}{r^2}\, r\,dr\,d\theta \\ &= \left(\int_0^{\pi/2} |\cos 2\theta|\,d\theta\right)\int_0^{\varepsilon}\frac{dr}{r} \\ &= +\infty, \end{aligned}$$

since $\int_0^{\varepsilon} dr/r$ diverges. Therefore $\iint_R |f|\,dA = \infty$, and condition (b) of Theorem 0.18 is not satisfied. (Condition (a) also fails: f takes both positive and negative values, so it is not nonnegative a.e.) The unequal iterated integrals from Example 0.41 are thus consistent with Theorem 0.18: the theorem simply does not apply.

(Wikipedia contributors 2024)

Further reading

- Kaplan (2022)
- Khuri (2003)
- Banner (2007)
- Larson and Edwards (2018)
- Miller (2016) The calculus review exercises and worked solutions across these pages and in Vector Calculus are adapted from this supplemental review chapter.
  - http://www.youtube.com/watch?v=xYzQL0TUtBA
  - http://www.youtube.com/watch?v=Ps2SBo_WjoE
- Grinberg (2017) (the rigorous foundations behind these results)
- Essence of Calculus is a 12-video YouTube playlist by Grant Sanderson (3Blue1Brown) that develops geometric and visual intuition for limits, derivatives, product and chain rules, implicit differentiation, integration, and Taylor series.

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